

AUGUST 2023



INCEPTION REPORT

Emission Trading System Piloting and Simulation in Vietnam

ABBREVIATIONS

A6SF	Article 6 Support Facility
ADB	Asian Development Bank
DCC	Department of Climate Change
ETP	The Southeast Asia Energy Transition Partnership
ETS	Emissions Trading System
GHG	Greenhouse gases
GIZ	The German Agency for International Cooperation
ICAP	International Carbon Action Partnership
JICA	Japan International Cooperation Agency
SPI - NDC	Support for Planning and Implementation of the Nationally Determined Contributions (NDC) in Vietnam
MARD	Ministry of Agriculture and Rural Development
MOC	Ministry of Construction
MOF	The Ministry of Finance
MOIT	Ministry of Industry and Trade
MONRE	Ministry of Natural Resources and Environment
MOT	Ministry of Transport
MPI	The Ministry of Planning and Investment
NDCs	Nationally Determined Contributions
NZ	Net-zero
SEP	Stakeholder Engagement Plan
South Pole	South Pole Carbon Asset Management
TOR	Terms of Reference
UNFCCC	United Nations Framework Convention on Climate Change
USAID	U.S. Agency for International Development
VETS	Viet Nam Technology Solutions Joint Stock Company
VNEEC	Energy and Environment Consultancy Joint Stock Company
VNPMR	Viet Nam Partnership for Market Readiness

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EXECUTIVE SUMMARY

To facilitate the transition towards a low-carbon economy, Viet Nam recognizes the importance of an Emissions Trading System (ETS) as a policy instrument to incentivize emissions reductions, promote the adoption of cleaner technologies, and create economic opportunities in sectors contributing to the transition. To this end, Viet Nam has adopted several policies establishing a legal framework for the development of a well-functioning carbon market and carbon trade exchange by 2027.

Still, the successful establishment and effective regulation of an ETS in Vietnam will require extensive capacity-building initiatives and activities aimed at increasing ETS literacy among relevant stakeholders, including policy makers in the country. The Assignment described in this Inception Report will therefore deliver a simulation training program to increase ETS literacy and collaboration between government authorities and the private sector.

It forms part of the Southeast Asia Energy Transition Partnership's (ETP) interventions aimed at developing the carbon market in Vietnam, and will be implemented in cooperation with the Vietnamese Ministry of Natural Resources and Environment (MONRE) from July 2023 to January 2025 by a consortium consisting of Energy and Environment Consultancy Joint Stock Company (VNEEC), South Pole Carbon Asset Management (South Pole), Viet Nam Technology Solutions Joint Stock Company (VETS), and international experts subcontracted by VNEEC.

The methodology and workplan of the Assignment envision ten specific tasks:

Task 1: Composing the Work Plan: This initial task will result in a comprehensive workplan that includes the project's objectives, activities, allocation of human resources, and delineated deliverables. It sets the overall direction of the project, ensuring that all activities are integrated and aligned with the project objectives.

Task 2: Organization of Inception Workshop: Hosting an inception workshop is the focus of the second task, which will entail sharing and discussing the detailed work plan, schedule, and deliverables of the Assignment, and gathering information and feedback from stakeholders. This task will ensure early engagement and a shared approach.

Task 3: Analysis and Consultations on Mitigation and Carbon Trading Readiness: Task three is intended to gather information and assess the readiness, awareness, needs, and gaps of regulated enterprises and other potential players in the carbon market, with a specific focus on the ETS. It will involve desk research and consultations, questionnaire-based surveys, data collection, compilation and analysis, and preparation of a readiness and awareness assessment report that includes a comprehensive list of potential participants for the subsequent training sessions.

Task 4: Development of Stakeholder Mapping and Engagement Plan: Drawing on the previous task, this task will yield a detailed stakeholder map based on agreed criteria and propose a stakeholder engagement plan designed to foster effective communication among all implementing partners, targeted groups, and stakeholders involved in the Assignment.

Task 5: Preparation of Carbon Simulation Tool: Task five encompasses the creation of a customized, policy-driven and user-friendly carbon simulation tool specifically designed for Viet Nam, including adjustment to the local context and testing ahead of broader roll-out.

Task 6: Provision of Web-based Carbon Simulation Platform: Task six consists of developing and continuously updating a web-based platform that will act as an update hub for the training and activities, serve as a knowledge hub, and offer a channel to increase public awareness regarding ETS and carbon pricing instruments.

Task 7: ETS Simulation Training Workshops and Forums: Through a comprehensive program of training and simulations, task seven will help facilitate learning and understanding of ETS principles and processes and disseminate knowledge and expertise on ETS. It will encompass pre-training activities, four in-person training sessions, and training materials and post-course assessment.

Task 8: Carbon Market Master Class for Key Government Officials: Task eight offers an in-depth on-the-job training course for key government personnel by identifying their capacity gaps and training needs and carrying out a study tour and master class in countries with relevant experiences or similar conditions to Vietnam.

Task 9: Technical Report on Simulations and Impact Assessments: Task nine consolidates the insights derived from the simulations and training in order to generate comprehensive recommendations for the implementation of the Vietnamese ETS and will consist of a technical report and high-level consultations.

Task 10: Delivery of Final Report and Workshops: The final task of the Assignment is intended to bring together the insights from the entire project, reflecting on notable achievements, lessons learned, and recommendations for the future direction of carbon trading in Viet Nam by way of a final report and closing workshop to present the outcomes and share insights with relevant stakeholders.

Stakeholder mapping and communication planning constitute important elements of the Assignment and will necessitate meticulous design of the network of stakeholders, engagement strategies, and communication protocols to ensure continuous information flow. The Assignment will also entail a donor coordination strategy that highlights the alignment with activities facilitated by other donors to streamline efforts and resources across different project phases.

In terms of project management, the Assignment will incorporate a logistical framework and strategies essential for effective execution. Key elements such as risk identification, mitigation strategies, and underlying assumptions will help guide the project in a manner that minimizes uncertainties and enhances predictability.

A gender impact assessment approach is embedded in the project, focusing on the identification of gender-sensitive indicators and addressing specific needs. This analysis aims to ensure that impacts on gender dynamics are fully understood, and necessary measures are taken to foster equality and inclusiveness.

Finally, the ETP Results-Based Monitoring Framework (RBMf) is an integral part of the Assignment, describing the methodology for monitoring results achieved from its implementation. It ensures alignment with the desired outcomes and guarantees that the project stays on course to meet its objectives.

1 INTRODUCTION AND PROJECT BACKGROUND

1.1 The role of the Emissions Trading System in transitioning towards a low-carbon economy

Viet Nam, a rapidly growing lower-middle-income country, faces significant vulnerability to the impacts of climate change. Recognizing the urgency of addressing this global challenge, Viet Nam has made a firm commitment to combat climate change by setting a bold target of achieving net-zero (NZ) emissions by 2050, aligning itself with the goals of many developed countries. This commitment serves multiple purposes for Viet Nam. Firstly, it signifies Viet Nam's contribution to the global efforts in mitigating climate change, recognizing the shared responsibility of all nations. Secondly, it is a strategic move to enhance Viet Nam's competitiveness in the global market due to the current climate change policies landscape globally as well as strengthening the sustainable development in Viet Nam.

To facilitate the transition towards a low-carbon economy, Viet Nam recognizes the importance of implementing the Emissions Trading System (ETS) as a crucial policy instrument. The ETS will play a significant role in incentivizing emissions reductions, promoting the adoption of cleaner technologies, and creating economic opportunities in sectors contributing to the transition. By implementing an effective ETS, Viet Nam aims to accelerate its progress toward a sustainable and resilient future while fulfilling its global climate commitments.

An ETS is a market-based instrument that can be used to reduce GHG emissions. ‘Cap and trade is a variant of an ETS where the government imposes a limit (cap) on total emissions in one or more sectors of the economy and, by extension, the emitting companies included in those sectors. At the end of each compliance period – typically one or more years – companies must either reduce their emissions (by effecting changes on-site) or secure sufficient allowances (or, if allowed, offsets) – from other companies and/or the government -- in amounts equal to each company’s emissions over that period of time. Over time, the jurisdiction-wide cap – and each company’s cap -- will decline at a rate that is sufficient to enable the jurisdiction to meet its obligations. Other critical elements of an ETS are featured in Box 1.

Box 1: Critical ETS elements

- **Accurate data:** The ETS is based on accurate and ever improving data.
- **Administration responsibilities:** ETS program administration responsibilities are clearly delegated to the central and various local governments.
- **Allocation method:** ETS administrators can distribute allowances for free, via government run sale/auction, or through some combination of methods. Free allowances may be distributed by considering a facility’s historical emissions rate, its comparative emissions intensity, and its exposure to competition from similar enterprises outside of the ETS. Allowances not distributed for free can enter the market via auctions.
- **Banking mechanism and registry:** An administrative mechanism that allows ETS participants to store allowances so that they can be held for use or offered for sale at a later date. The existence of an emissions bank promotes investments in measures that reduce emissions and free up allowances. This feature may help encourage enterprises to over-comply and financial institutions to provide funding for such initiatives.

- **Capacity building:** To help ensure program success necessary stakeholder capacity building efforts should be implemented on a continuing basis.
- **Common requirements:** ETS programs are structured around a common core of requirements that apply to all entities within each category, including those applicable to ETS administrators, enterprises, market operators, and other stakeholders.
- **Compliance flexibility:** An ETS will also provide the flexibility to achieve emissions reduction in a variety of ways, for example: one or multiple sources on-site at a given enterprise, through another source within the control of an enterprise, or by paying another enterprise covered by the program to make the reductions instead.
- **Emissions allowances or quotas:** The currency of an ETS, allowances, provide enterprises with the permit to emit a defined quantity (often in tons) of emissions.
- **Emissions reduction deadline:** The cap will include interim (such as annual) deadlines by which time programmatic and enterprise-specific emission reductions must be achieved.
- **Enforcement:** There should exist a means by which the ETS administrator can impose meaningful consequences upon ETS participants (enterprises and traders) that emit more than allowed or act in other ways that are contrary to the ETS rules. Such consequences may include fines (flat or structured to correlate with recently transacted prices or the magnitude of the exceedance and nature of the violation), the forfeiture of allowances that have been (or would have been) issued in subsequent years of the ETS, or incarceration. Depending upon the ETS and the nature of the transgression, the enterprise or one or more responsible individuals may be held liable.
- **Enterprise-specific caps/limitations and obligations:** Once caps/limits are established for the entire ETS, enterprise specific emission obligations (caps/limits) are defined. In this way, each enterprise is allocated its proportional share of the cap (as defined by the ETS administrator). Further, specific operating requirements (and limits) should be assigned to each enterprise.
- **Equity:** Care must be taken to help ensure that emission reduction obligations and opportunities are equitably imposed on those that are required to reduce emissions.
- **Goal:** An ETS should be designed to achieve a central goal (e.g., achieving a particular emission reduction by a defined date). Secondary goals (e.g., increasing green jobs, generating revenue for the government, or advancing low carbon technologies) may also be included but should not undermine the primary goal.
- **Fund management:** An ETS, particularly one that features an auction and/or which includes the collection of fines, will collect monies. The administrator should be provided with guidance (and oversight) as to how such monies are collected and disbursed. In general it is a good idea to ensure that all (or a portion) of the funds are used to support objectives that are consistent with those underlying the ETS itself – e.g., facilitate a low carbon fuel transformation, support/encourage mass transit alternatives, address environmental justice concerns, and mitigating the adverse consequences of climate change that have been/are/will be felt by at risk communities and current and future generations.
- **Legal foundation:** Authority to develop and administer the ETS is derived from a sound underlying legal foundation that eliminates uncertainty amongst stakeholders that the administering entity has the authority to implement the ETS and the enterprises must comply with the requirements related thereto.
- **Liquidity:** Liquidity is a measure of how easy it is to convert an asset to cash or how rapidly the asset can be sold. In carbon markets, greater liquidity signifies a healthier

market because it means that companies are able to buy and sell emissions allowances in a timely manner to fulfil compliance obligations and manage risk.

- **Long-term decisions:** The ETS includes measures that encourage enterprises to make decisions that allow for investments and compliance strategies that may be implemented over the short or long term.
- **Market:** A marketplace provides a means by which quotas and qualifying offsets can be transacted. Such markets can run through government-sponsored auctions, exchanges (private or government sanctioned), over the counter (directly between two parties), or by another means as dictated by the needs of the ETS and its participants. Generally, the operators and participants in such a marketplace are subject to government issued regulations and oversight.
- **Monitoring, reporting, verification (MRV):** An ETS must include a means by which actual emissions can be accurately monitored, reported to the enterprise and the government, and verified by the government. A primary objective is to ensure that emissions, allowances, and reductions are consistent with the ETS requirements, are accurately accounted for, and not double counted.
- **Offsets:** Emission offsets represent emission reductions that are derived from actions taken by entities that operate emission sources that are not included in the ETS. Such reductions are created in accordance with government prescribed protocols. Under certain circumstances, emission offsets can be used in lieu of allowances. The presence of offsets can serve to increase the supply of allowances and thereby moderate allowance prices. It is common for an ETS to restrict the supply of offsets (e.g. by limiting the types of sources that are eligible to generate offsets, creation methods, and geographic areas from which they can be sourced, or methodologies that can be used), limit the maximum quantity that can be used (such as 5-10% of an enterprise's compliance obligation), and prohibit the use of emission reductions which may be associated with shifting demand. The ETS administrator may elect to give preference to offsets that also deliver meaningful co-benefits (e.g., are created locally, facilitate a clean fuel transition, support women and minority owned businesses, support ecosystem and/or rainforest preservation, etc.). Finally, to help avoid leakage, the ETS may require that offsets be created, accounted for, and managed on a jurisdictional basis.
- **Program-evaluation/adjustment:** The ETS should be subject to regular program-wide reviews with the aim of assessing its performance and recommending changes aimed at ensuring that key objectives are met.
- **Program-wide emissions cap/limit:** The cap/limit represents the amount of emissions that can be released by all of the enterprises that are included in the program. Generally, the cap/limit establishes a quantity of emissions that can be allocated at the start of the program, and a reduction that must be achieved in order to realize a health-based or environmentally-based standard.
- **Reconciliation:** At the end of each compliance period (such as each year) enterprises must true-up their compliance status. Those with emissions that exceed the quantity of retired quotas must secure and retire additional allowances. Those that do so may be judged to be in compliance with a central requirement of an ETS.
- **Reserve:** A quantity of allowances may be held by the ETS administrator and released as per prescribed protocol (such as to fund auctions or to enhance or moderate volatility) or augmented by withdrawing previously distributed or acquired allowances from individual enterprises.

- **Risk management:** Market participants should be allowed to develop appropriate tools that can be used to manage/hedge risk.
- **Scarcity:** Each enterprise covered by the ETS may be provided with a quantity of allowances that, in the aggregate, declines over time such that the enterprise must either reduce emissions or acquire allowances from other participants. Scarcity can also be influenced through limits that are imposed on the quantity of allowances that can be banked and/or through the injection of allowances via auctions.
- **Scope and thresholds:** The ETS will define criteria (typically type of industry and quantity of emissions) that are used to select which enterprises are included in the ETS. Such criteria will be consistent with the nature of the challenge and the resources that are needed to administer, support, and be subject to the ETS.
- **Stakeholder involvement:** The ETS should be developed, administered, and evaluated with the involvement of relevant stakeholders – particularly those from relevant government agencies, affected sources, civil society, ETS service providers, offset providers, and the public.
- **Transparency:** The ETS requirements should be clear and understood by all stakeholders. So too should the performance of the enterprises that must comply with the requirements and the ETS administrators. Transparency should also extend to the market wherein the quotas and offsets are transacted.

To comply with the emissions limits and minimize their costs, enterprises may adopt a carbon portfolio that features a variety of emission reduction strategies. These strategies may include improving energy efficiency, transitioning from fossil fuels to renewable energy sources, and implementing low-carbon technologies in their production processes. Hence, a well-designed, implemented, and administered ETS will drive affected businesses to invest in emission reduction measures and stimulate innovation in low-carbon technologies. On a jurisdiction-wide basis, this can drive the transition towards a low-carbon economy and foster the development of new technologies that contribute to overall emissions reduction within its jurisdiction.

1.2 State of the ETS globally and in Viet Nam

1.2.1 ETS around the world

As of March 31, 2023, according to World Bank¹, a total of 36 ETSs have been implemented globally. These ETS cover 12 national jurisdictions, indicating the extent of their implementation at the national level. These systems collectively regulate approximately 8.91 GtCO₂e, which accounts for approximately 17.64% of global GHG emissions. These figures demonstrate the growing recognition and adoption of ETS as a policy instrument to address and regulate GHG emissions on a significant scale.

¹ https://carbonpricingdashboard.worldbank.org/map_data

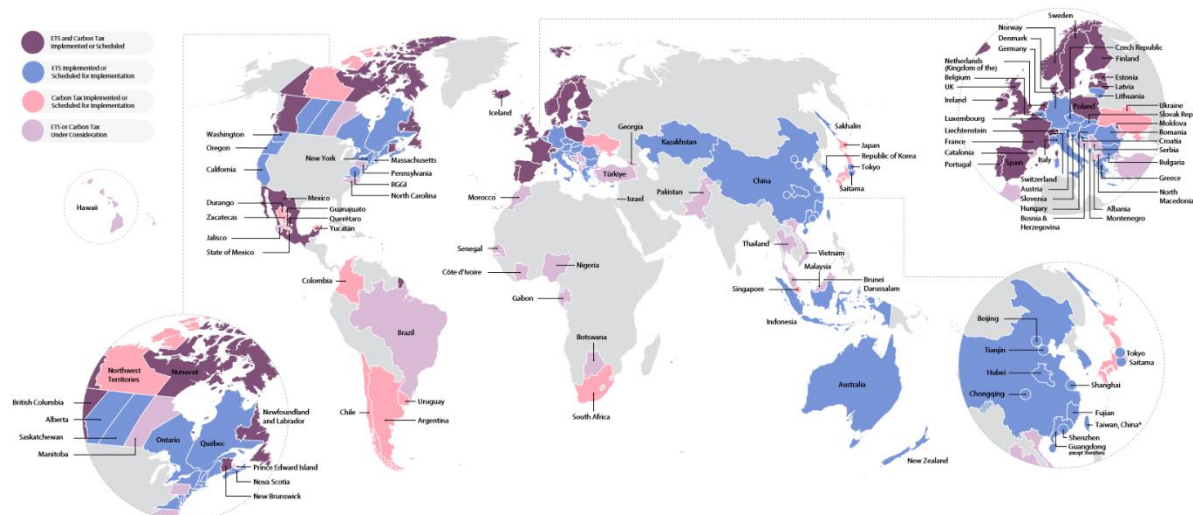


Figure 1: Map of Carbon taxes and ETS (World Bank, 2023)

1.2.2 The development of the Viet Nam carbon market

In November 2020, Viet Nam's revised "Law on Environmental Protection" was promulgated, providing for the first time the organization and development of the domestic carbon market (Article 139).

The Government of Viet Nam took a significant step on January 7, 2022, by issuing Decree No. 06/2022/ND-CP, titled "Regulations on mitigation of GHG emissions and protection of the ozone layer." This decree provides detailed guidelines on two aspects: Article 91, focusing on the reduction of GHG emissions, and Article 139, focusing on the formation and development of the carbon market as outlined in the Law on Environmental Protection 2020. Chapter 2 of the Decree specifically addresses the development of Viet Nam's domestic carbon market, encompassing regulations, and provisions of the market.

The development roadmap of Viet Nam's domestic carbon market is regulated as in the figure below:

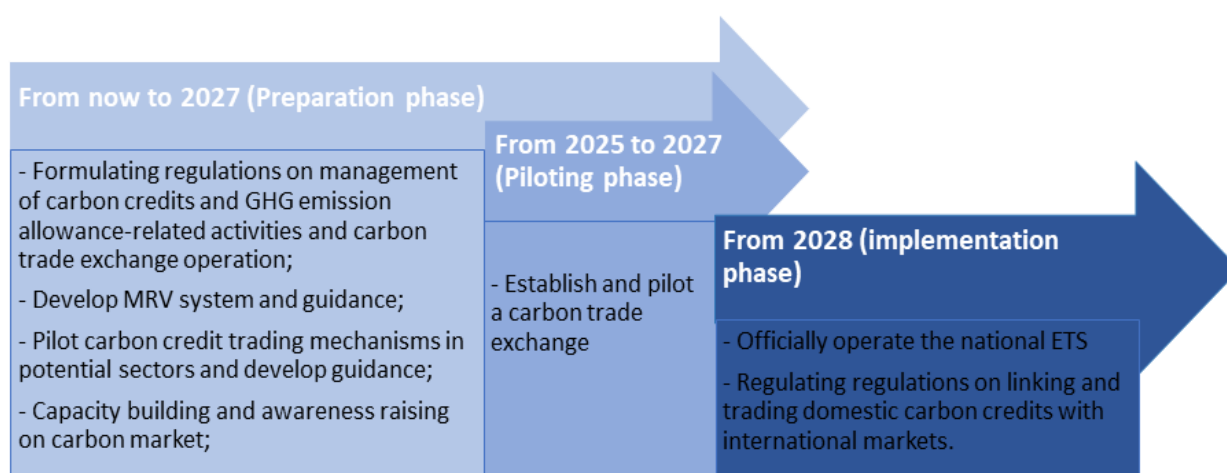


Figure 2: Roadmap of the domestic carbon market of Viet Nam²

On January 18, 2022, the Prime Minister issued Decision 01/2022/QĐ-TTg “on the list of sectors and facilities emitting GHG that must carry out an inventory of GHGs”. This decision mandates a total of 1,912 GHG emitters to conduct GHG inventories or participate in Viet Nam's domestic carbon market.

These policies establish a legal framework for the development of the domestic carbon market and highlight the importance of establishing an ETS in Viet Nam. The successful establishment and effective regulation of the ETS in Viet Nam require extensive capacity-building initiatives and activities aimed at increasing ETS literacy among relevant stakeholders, including policy makers in the country.

Therefore, an ETS simulation training program to increase ETS literacy and collaboration between government authorities and the private sector are essential factors for the successful establishment and operation of the ETS. These efforts will contribute not only to paving the way for an ETS roll-out but also help Viet Nam to achieve the NZ targets.

2 SCOPE OF SERVICES

This Assignment is part of the Southeast Asia Energy Transition Partnership’s (ETP) interventions aiming to develop the carbon market in Viet Nam in cooperation with the Ministry of Natural Resources and Environment (MONRE) and the Ministry of Finance (MOF). With MONRE, ETP will support the Department of Climate Change (DCC) to: pilot and model ETS simulation, carbon pricing design, pilot voluntary labelling program and assessment of the opportunities to link the domestic ETS with the international ETS. With MOF, ETP will support the Legal Department identifying the legal, financial and infrastructure gaps for the carbon trade exchange development and filling the gaps through a series of follow-up technical assistance and capacity building activities. The ultimate goal of these interventions is to support Viet Nam to establish a well-functioning carbon market and carbon trade exchange by 2027.

The work required under this Assignment include 10 specific tasks:

- Task 1: Conducting an Inception report

² Source: VNEEC based on Degree 06/2022/ND-CP

- Task 2: Organizing an Inception workshop
- Task 3: Conducting an analysis, surveys and consultation events on readiness and awareness of enterprises and facilities on GHG emission/mitigation and carbon market.
- Task 4: Developing a detailed stakeholder mapping and engagement plan to approach and engage the targeted groups.
- Task 5: Providing access to a carbon simulation tool suitable for use for this Assignment that is tailor-made for Viet Nam in terms of language, currency, years, duration, offset, price collar, banking, and shocks.
- Task 6: Providing a web-based carbon simulation platform.
- Task 7: Organizing a series of ETS simulation trainings/workshops/forums.
- Task 8: Planning a study tour on ETS governance and experience sharing.
- Task 9: Conducting a technical report summarizing results of the simulation and impact assessment.
- Task 10: Final report and final workshops.

Through an ETS simulation tool and a hands-on learning platform for various participants and stakeholders involved in the ETS. Training sessions on the ETS using the tool and the platform will be delivered to different ETS participants and players to raise their awareness and readiness regarding the future ETS in Vietnam. In addition, the Consultant will develop concrete policy recommendations for designing an ETS in Viet Nam based on the analysis of the lessons learnt and impacts of the ETS operation on energy transition. The Assignment will directly support the introduction of a domestic ETS as a carbon pricing instrument by 2027.

As per the Contract signed between UNOPS and the Consultant, the Assignment is scheduled to take place over a period of 18 months, starting from July 2023 and concluding in January 2025.

The implementation of the Assignment will be carried out by a consortium consisting of Energy and Environment Consultancy Joint Stock Company (VNEEC), South Pole Carbon Asset Management (South Pole), Viet Nam Technology Solutions Joint Stock Company (VETS), and international experts subcontracted by VNEEC. Within the consortium, VNEEC will take the overall coordination and management of the Assignment. The Consultant/Consortium combines a strong national qualification and experience with international experience and lessons to deliver the Assignment to the highest quality.

The primary beneficiaries of this Assignment include MONRE as a whole, with a specific focus on DCC, as well as the line ministries and agencies responsible for the design and operation of the future ETS in Viet Nam. Additionally, the main beneficiaries encompass GHG emitters who are required to participate in the future ETS. Annex I provides a list of other stakeholders who can be considered as beneficiaries under this Assignment.

3 METHODOLOGY AND WORKPLAN

3.1 Task 1: Composing the Inception Report and Project Workplan

3.1.1 Objectives

The primary aim of the Inception phase is to establish a shared understanding between the Consultant and the Clients regarding the key aspects of the Assignment. During this phase, discussions will be held to reach an agreement on the approach and methodology for accomplishing the required tasks and any necessary adjustments. This includes updating the tentative timeline to ensure its alignment with the agreed-upon approach. The Inception phase sets the foundation for successful collaboration and sets the stage for the subsequent stages of the Assignment.

3.1.2 Activities

3.1.2.1 Activity 1.1: Conducting Inception kick-off meeting

On 13 July 2023, following the signing of the Contract between UNOPS and VNEEC, VNEEC took the initiative to prepare sub-contracting agreements with other consortium members, including South Pole and VETS. This step was crucial to ensure the timely onboarding of experts for the Assignment's implementation.

On 31 July 2023, the Consultant organized a kick-off meeting with the ETP. The primary objectives of this meeting were to introduce the Consultant team, prepare the Inception Report and discuss subsequent steps. The Consultant presented an overview of the Assignment, timeline, team management, and the overall approach and methodology for executing task 3. An overview of the simulation tool is also provided. Detailed minutes of the kick-off meeting between the Consultant and ETP are provided in Annex IV.

3.1.2.2 Activity 1.2: Preparing inception report and updating work plan

Taking into account the Terms of Reference (TOR) and the discussions held during the kick-off meetings with the Client and relevant key stakeholders, an Inception Report has been prepared and submitted. The Inception Report establishes a framework for the consistent implementation of the Assignment from initiation to completion.

To ensure a collaborative and inclusive process, close consultation and engagement with the DCC and the ETP will be maintained. This is done to seek final approval before proceeding with the subsequent activities of the Assignment. By involving these key stakeholders, the implementation process can benefit from their expertise and insights, leading to a more effective and well-informed execution.

3.1.3 Allocation of human resources

Details of human resources of this Task are provided below:

Table 1: Details of human resources for Task 1 (W1-W2, W3-W6, 01 Aug-10 Aug 2023, 20 Aug - 10 Sep 2023)

No	Personnel and total time inputs	Person days
1	Team Leader/Climate Change Policy Expert: Dang Hong Hanh	5
2	National GHG Inventory/Mitigation Expert: Tran Minh Tuyen	1
3	International GHG Inventory/Mitigation Expert: Roxanne Tan	3
4	Carbon Market Simulation Expert: Josh Margolis	3
5	Stakeholder Engagement Expert 1: Luu Minh Nghia	2
6	International Capacity and Training Expert 1: Karolien Casaer-Diez	3
7	International Capacity and Training Expert 2: Michael Mehling	2
9	National Capacity and Training Expert 1: Nguyen Tien Hai	1

10	International Carbon Pricing Policy Expert: Victor Escalona Gomez	2
11	Survey Data Collection Expert: Dao Xuan Hoach	3
12	IT Expert: Duong Chi Cong	3
13	Supporting experts (including administration and translation)	

3.1.4 Deliverables

The Inception Report in English and Vietnamese.

Together with the Inception Report, the following are to be submitted:

- A public facing, publishable Executive Summary (no more than 2 pages) in professional English must be submitted with each deliverable.
- A public facing, catchy PowerPoint presentation highlighting key information must be submitted with each deliverable.

3.2 Task 2: Organizing an Inception Workshop

3.2.1 Objectives

The primary objective of this task is to present the Assignment and gather information and feedback from stakeholders. During this process, the detailed plan, schedule, and deliverables of the Assignment will be shared and discussed. The aim is to gather input and feedback from stakeholders regarding these aspects. Additionally, this task seeks to explore potential collaboration opportunities with other activities to prevent overlap and ensure effective coordination. By engaging stakeholders and seeking their support and collaboration, the implementation of the Assignment can benefit from their insights and involvement, leading to a more successful outcome.

3.2.2 Activities

3.2.2.1 Activity 2.1: Preparing for the Inception Workshop

In consultation with DCC and the ETP, the Consultant will prepare the Agenda and propose a list of participants for the inception workshop. The specific key stakeholders involved in this process, along with their contact information and roles and responsibilities, have been meticulously identified and outlined in Annex I. By engaging with the DCC, ETP, and other relevant stakeholders, the Consultant aims to ensure a comprehensive and inclusive consultation process that gathers valuable input and promotes collaboration throughout the Assignment's implementation.

3.2.2.2 Activity 2.2: Organizing the Inception Workshop

The Consultant, with approvals from MONRE/DCC and ETP, will organize the Inception Workshop with the participation of the key stakeholders identified in the Inception Report.

3.2.3 Allocation of human resources

Details of human resources of this Task are provided below:

Table 2: Details of human resources for Task 2 (W1- W6, 01 Aug -10 Sep 2023)

No	Personnel and total time inputs	Person days
1	Team Leader/Climate Change Policy Expert: Dang Hong Hanh	5
2	National GHG Inventory/Mitigation Expert: Tran Minh Tuyen	1

3	International GHG Inventory/Mitigation Expert: Roxanne Tan	4
4	Carbon Market Simulation Expert: Josh Margolis	3
5	Stakeholder Engagement Expert 1: Luu Minh Nghia	4
6	International Capacity and Training Expert 1: Karolien Casaer-Diez	4
7	International Capacity and Training Expert 2: Michael Mehling	2
8	National Capacity and Training Expert: Nguyen Tien Hai	3
9	International Carbon Pricing Policy Expert: Victor Escalona Gomez	1
10	Survey Data Collection Expert: Dao Xuan Hoach	5
11	IT Expert: Duong Chi Cong	5
12	Supporting experts (including administration and translation)	

3.2.4 Deliverables (Milestone 1)

- The Final Inception Report in English and Vietnamese submitted to ETP and DCC
- Inception Workshop minutes and proceedings

If it is requested, the followings are to be submitted:

- A public facing, publishable Executive Summary (no more than 2 pages) in professional English must be submitted with each deliverable.
- A public facing, catchy PowerPoint presentation highlighting key information must be submitted with each deliverable.

3.3 Task 3: Conducting an analysis, surveys and consultation events on readiness and awareness of enterprises and facilities on GHG emission/mitigation and carbon market

3.3.1 Objectives

The primary goal of this task is to gather information and assess the readiness, awareness, needs, and gaps of regulated enterprises under Decision No.01/2022/QĐ-TTg, as well as other potential key players in the carbon market, with a specific focus on the ETS. Subsequently, a comprehensive list of potential participants for the training sessions will be compiled.

3.3.2 Activities

3.3.2.1 Activities 3.1: Conducting the desk review and consultations on the ETS (non-emitter entities)

To identify the gap on awareness and readiness of relevant stakeholders regarding the ETS, desk review and a series of consultation with relevant stakeholders will be conducted focusing on:

- International and domestic climate policy related to the development of the ETS in Viet Nam.
- Previous relevant training activities in Viet Nam and internationally focusing on approaches, structures, content, and results of the training.
- International experience in identifying ETS participants.

The desk review will encompass a diverse array of publications and training materials, encompassing both international and domestic climate policies. This includes an examination of previous training initiatives conducted by the International Carbon Action Partnership (ICAP), as well as ETS simulations carried out in Viet Nam and other countries. Furthermore, insights from the Viet Nam - Partnership for Market Readiness will be utilized to establish an initial framework for the training component of this Assignment.

To ensure the training framework aligns with the specific capacity gaps and requirements for designing and implementing the ETS in Viet Nam, consultations will be undertaken with key policy makers with a focus on DCC under MONRE and the Legal Department under MOF as these are the most relevant government agencies responsible for the establishment of the ETS in Viet Nam. Other consultations with other government agencies will be identified after the consultations with DCC and the Legal Department of MOF. These consultations will aim to identify the actual capacity gaps and needs related to the ETS's design and operation from the ETS administrator's point of view. The insights gathered from these consultations will be carefully considered and integrated into the training framework, as well as utilized for the development of an awareness and readiness report.

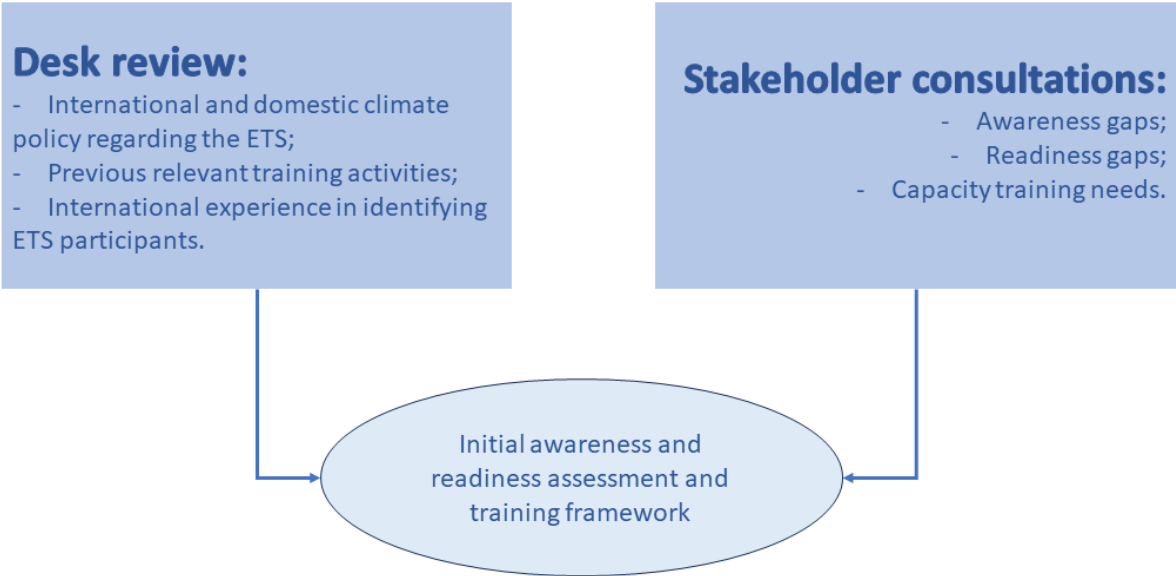


Figure 3: Workflow and expected results under activity 3.1

3.3.2.2 Activity 3.2: Design the survey questionnaire

Once the initial awareness and readiness assessment and capacity training framework are identified, a set of survey questions will be developed. The questionnaires will be designed to encompass primary sections, including **respondent's profile, ETS baseline knowledge, readiness and others**. These sections will seek to assess the preparedness of the participants and their level of understanding regarding the ETS through a range of carefully constructed questions. To ensure the cooperation of the enterprises and a secure response rate, the length of the questionnaire will be controlled, and questions are intended to be just enough to assess awareness and readiness of the stakeholders regarding the ETS. The questionnaire will be designed in both English and Vietnamese. However, only the Vietnamese version will be used for the survey.

The awareness part will cover, but is not limited to, the following topics:

- Understanding and perceptions of current carbon pricing mechanisms.
- Understanding and perceptions of government regulation and their obligations regarding the future domestic carbon market in Viet Nam.
- Understanding and perceptions of climate change and its impact on their operation.

The readiness part will cover, but is not limited to, the following aspects:

- Energy saving/efficiency improvement potential of the company.
- Current energy consumption/Operating capacity of the emitter.
- The capacity and years of experience of the company regarding measurement and reporting of energy consumption and/or GHG emissions, GHG emissions inventory, GHG mitigation best practices.
- Company specific short- and long-term GHG emissions mitigation plan.

The design of the questionnaires will be conservatively constructed, drawing upon the consultants' extensive domestic and international experience in the field. Furthermore, the questionnaires will be incorporated by the insights obtained through Activity 3.1 with key policymakers and relevant stakeholders conducted in the aforementioned activity.

The main purpose of the questionnaires will be to pinpoint the specific requirements of the private sector regarding their involvement in the future ETS. This crucial information will form the basis for creating customized training content that directly addresses these needs. By doing so, the project aims to enhance the private sector's comprehension and preparedness. The data collected through the survey will also contribute as inputs to the readiness and awareness assessment report.

Furthermore, the questionnaire will include a company profile section to collect background information from the emitters, aiming to provide a comprehensive profile of the targeted group. The section will focus on the following aspects:

- **Energy consumption:** Gathering data on the energy consumption patterns of the emitters, such as the types of energy sources/feedstocks utilized.
- **Business profile:** Obtaining information about the nature of the emitter's business activities and products, their industry sector, size, operational scale, and production processes.
- **Geographic location:** Identifying the geographic location of the emitters.

By collecting this additional background information, the survey aims to provide a comprehensive understanding of the emitters and their specific contexts, contributing to the selection progress of the potential players to participate in the training.

3.3.2.3 Activity 3.3: Determine the scale and coverage of the surveys

In order to establish a thorough and aligned database in accordance with the carbon market development framework in Viet Nam, the Consultant will conduct surveys among the identified GHG emitters who are obligated to carry out GHG inventory as stated in Decision No.01/2022/QĐ-TTg, or among the GHG emitters who might participate in the forthcoming domestic carbon market in Viet Nam.

However, the number of entities stipulated in Decision No.01/2022/QD-TTg is large (1,912 entities) and it would not be possible to survey all entities for the purposes of this Assignment. In addition, during the pilot and early phases of the ETS, not all emitters are likely to be allocated emissions allowances or regulated under the ETS. Currently, no criteria or information has been published regarding the selection of emitters to be participated in the Viet Nam’s ETS. Furthermore, based on the Consultant’s initial analysis, there is an uneven distribution of GHG emitters in various sub-sectors within the primary four sectors that exhibit distinct emission profiles and structures.

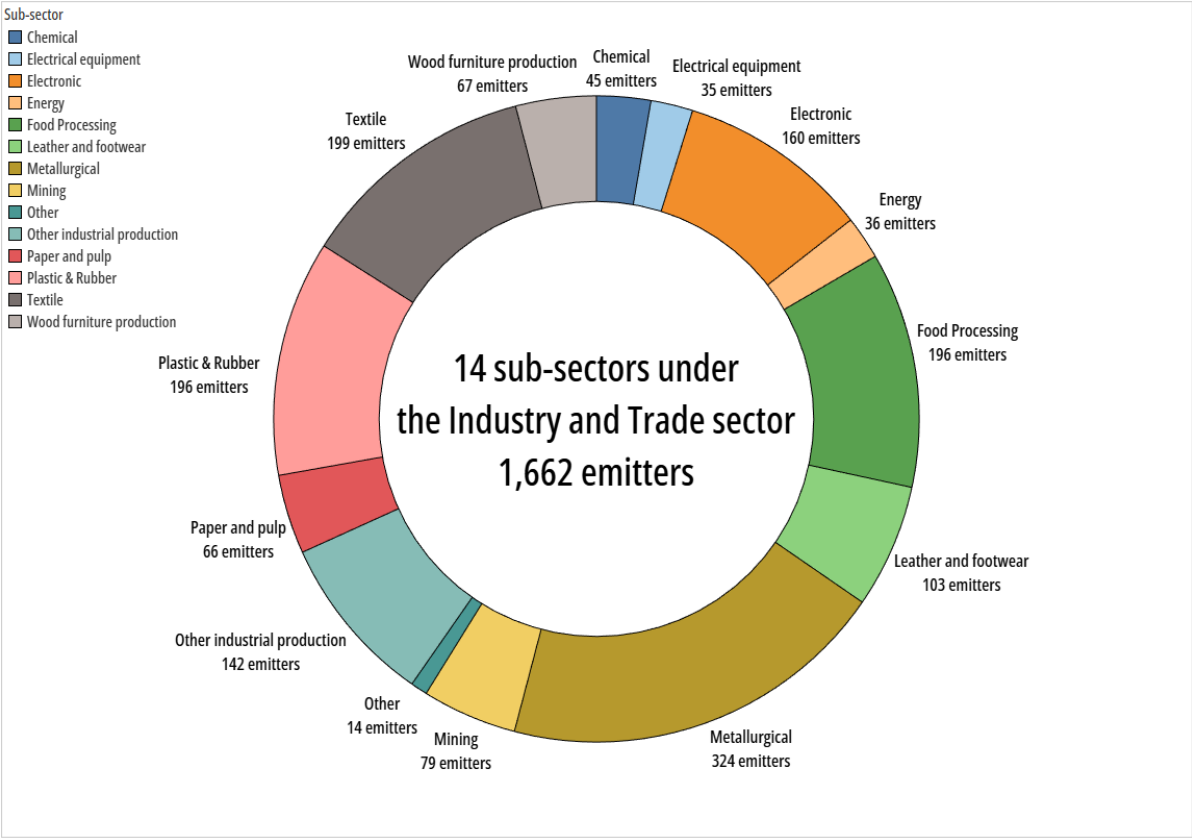


Figure 4: The number of emitters in 14 sub-sector of the Industry and Trade sector based on Decision No.01/2022/QD-TTg

Source: The Consultant analysis based on based on Decision No.01/2022/QD-TTg

Considering the training's efficacy, it is crucial to limit the number of participating emitters. Consequently, the Consultant’s approach is to identify big emitters that should be participating in the training, while guaranteeing that representation from every sector within the forthcoming ETS in Viet Nam is achieved.

- **Step 1:** Using data readily available to consultant, estimate the corresponding historical GHG emissions of each emitter that has been listed in Decision No.01/2022/QD-TT and additional data on their energy consumption and operating capacity³; To ensure a conservative approach, only Scope 1 and Scope 2 emissions of these emitters will be considered in the estimation.

³ It is assumed that the most updated data of Decision No.01/2022/QD-TT will be provided by DCC for further analysis.

- **Step 2:** Based on the historical GHG emissions obtained from the previous step, the average GHG emissions for each sub-sector will be determined in order to establish sub-sectoral thresholds. Each sub-sector threshold will be determined as the average emissions of companies in a specific sub-sector. These thresholds will serve as reference points to identify big emitters within their respective sub-sectors.
- **Step 3:** The emitters with historical GHG emissions above and close to the sub-sectoral thresholds, which indicate significant levels of emissions, will be categorized as "big emitters". Therefore, the survey will primarily focus on targeting these big emitters to gather comprehensive data and insights related to their emissions, operational practices, and potential mitigation measures as well as their readiness and awareness regarding the future ETS in Viet Nam.

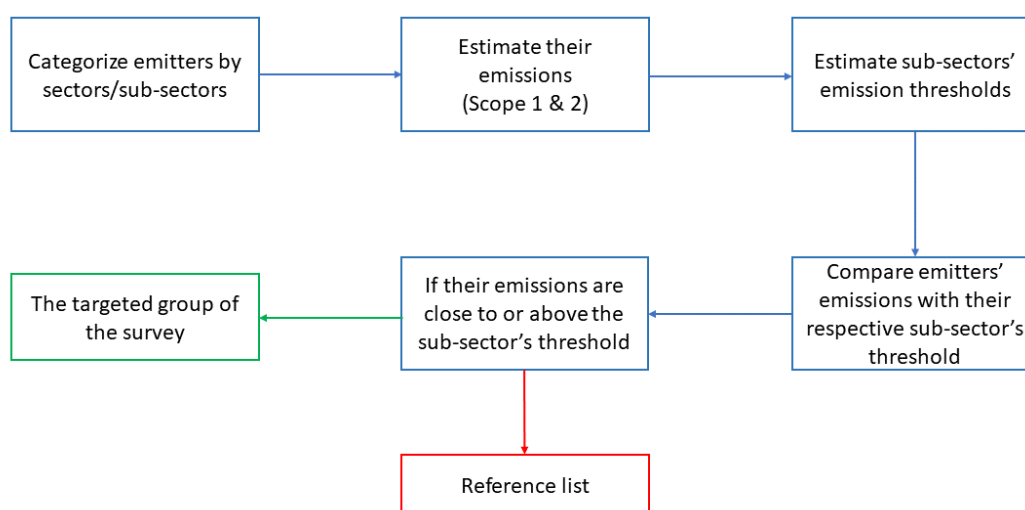


Figure 5: Workflow to determine the scope of the survey

The key participants of the training courses will be selected based on the targeted group of the survey given that they are among the largest emitters in the country who will be potentially targeted first in the pilot and early phase of ETS. However, the training courses will not exclude the group of smaller emitters, so the emitters listed in the reference list will be considered to select and invite to the training when suitable.

3.3.2.4 Activity 3.4: Data collection, compilation, and analysis

The survey will be administered through both online and postal channels. Additionally, if the information gathered from the desk review and consultations needs further clarification or additional data, some relevant stakeholders who were consulted in Activity 3.1 may also receive the survey. To ensure a minimum response rate of 20% (with good quality and informative responses), the Consultant will proactively follow up with the recipients after sending out the questionnaires and use the events, public media to get the interest from targeted groups (e.g. the Inception Workshop will be served as a platform to introduce the survey and get the attention/support and public).

The collected questionnaire results will be compiled within one month from the initiation of the task. Subsequently, the Consultant will evaluate the answer sheets of each potential participant. In conducting the questionnaires, attention will be given to safeguarding the confidentiality of the respondents' provided information. The Consultant commits in the questionnaire form to use the

collected information within and for the purposes of the Assignment only. It is important to note that the survey will be carried out under the guidance and endorsement of DCC, MONRE, and ETP, providing a framework for the Consultant to conduct the survey in alignment with the designated authorities.

The collected data will be stratified to separate the data into groups and categories so that patterns can be seen. Visualizations, charts and tables will be plotted to be included in the report.

3.3.2.5 Activity 3.4: Preparing a readiness and awareness assessment report and proposing a list of potential players to participate in the training

The data and information collected from activity 3.1, in combination with the survey results, will be vital in generating a comprehensive readiness and awareness assessment report. This report will offer valuable insights into the preparedness, awareness, needs, and gaps of potential key participants who will be participating in upcoming training courses. Furthermore, the readiness and awareness assessment report will serve as inputs for the DCC, providing a foundation for future activities in Viet Nam to ensure the effective preparation and implementation of the future ETS. Moreover, the survey inputs will be utilized to make necessary adjustments that can be incorporated into the simulation at a later stage.

3.3.3 Allocation of human resources

Details of human resources of this Task are provided below:

Table 3: Details of human resources for Task 3 (W5- W16, 01 Sep -30 Nov 2023)

No	Personnel and total time inputs	Person days
1	Team Leader/Climate Change Policy Expert: Dang Hong Hanh	8
2	National GHG Inventory/Mitigation Expert: Tran Minh Tuyen	10
3	International GHG Inventory/Mitigation Expert: Roxanne Tan	6
4	Carbon Market Simulation Expert: Josh Margolis	1
5	Stakeholder Engagement Expert 1: Luu Minh Nghia	10
6	International Capacity and Training Expert 1: Karolien Casaer-Diez	6
7	International Capacity and Training Expert 2: Michael Mehling	1
8	National Capacity and Training Expert: Nguyen Tien Hai	10
9	International Carbon Pricing Policy Expert: Victor Escalona Gomez	6
10	Survey Data Collection Expert: Dao Xuan Hoach	24
11	IT Expert: Duong Chi Cong	3
12	Supporting experts (including administration and translation)	

3.3.4 Deliverables

- Survey questionnaire approved by ETP and DCC.
- Assessment report on awareness, readiness, and capacity training needed.

If it is requested, the followings are supposed to be submitted together:

- A public facing, publishable Executive Summary (no more than 2 pages) in professional English must be submitted with each deliverable.
- A public facing, catchy PowerPoint presentation highlighting key information must be submitted with each deliverable.

3.4 Task 4: Developing a detailed stakeholder mapping and engagement plan to approach and engage the targeted groups

3.4.1 Objectives

The objectives of this task are to conduct a stakeholder mapping and engagement plan to approach and broadly engage the targeted groups through the training sessions under this Assignment.

3.4.2 Activities

3.4.2.1 Activity 4.1: *Developing a detailed stakeholder list/map*

The Consultant will analyze and categorize the results obtained from the previous task to compile a list of potential participants of the training sessions in Task 7. The selection process for participation in the training will prioritize the following criteria:

- Having a substantial emissions profile, making them key representatives of their respective sub-sectors.
- Demonstrating high interest in the ETS.
- Representing diverse geographic locations across the country.

In addition, an extended list of participants will be proposed, utilizing the comprehensive stakeholder mapping results. This expanded list will encompass players from various sectors, including:

- **Key government officials** from line ministries and agencies responsible for overseeing sectors involved in the development and operation of the future ETS.
- **ETS service providers and offset developers** who may play crucial roles in supporting the ETS implementation.
- **Other stakeholders** engaged in the development and operation of the domestic carbon market in Viet Nam. This may include legal entities, financial institutions, research organizations, NGOs, and NPOs, among others.

By incorporating these additional stakeholders, the subsequent training initiatives can benefit from a wider range of perspectives and expertise, fostering a more inclusive and collaborative approach to support the development and operation of the ETS in Viet Nam.

Once the potential list of participants is identified, the Consultant will conduct a thorough stakeholder mapping to categorize them by interest (how much they are impacted by or want to be involved in the Assignment) and influence or power (how much they can impact the Assignment). This activity will utilize and update information and findings obtained from the Inception workshop and other previous activities.

3.4.2.2 Activity 4.2: *Developing a stakeholder engagement plan*

The Consultant will formulate a stakeholder engagement plan designed to foster effective communication among all implementing partners, targeted groups, and stakeholders involved in this Assignment. The goal is to ensure active engagement throughout the entire timeline of the Assignment, maximizing the outcomes of the project. The Consultant will compose the following initial structure for the engagement plan. The initial stakeholder engagement plan outline that will include a communication plan is provided in Annex II and III.

Box 2: Initial structure of the stakeholder engagement plan

- **Introduction:** Provide a concise overview of the project and outline the objectives of the engagement plan.
- **Stakeholder identification and approach:** Detail the roles and strategies for engaging the targeted groups and stakeholders identified in Activity 4.2.
- **Information disclosure strategies:** Specify the information and content to be disclosed, along with the formats, platforms, and frequencies of disclosure.
- **Communication strategy:** Propose a communication strategy that promotes project activities publicly, emphasizing the engagement and commitment of the targeted groups. For instance, participants who complete the full training programs will receive a certificate signed by the DCC and ETP.
- **Review of comments and feedback:** Explain the process for gathering and reviewing comments and feedback, with a commitment to report back to stakeholders on final decisions and provide a summary of how comments were incorporated to continuously improve the simulation.
- **Management functions and responsibilities:** Describe how stakeholder engagement activities will be integrated into the Assignment's management system and specify the staff dedicated to implementing and managing the Stakeholder Engagement Plan.

3.4.3 Allocation of human resources

Details of human resources of this Task are provided below:

Table 4: Details of human resources for Task 4 (W13- W16, 01 Nov -30 Nov 2023)

No	Personnel and total time inputs	Person days
1	Team Leader/Climate Change Policy Expert: Dang Hong Hanh	8
2	National GHG Inventory/Mitigation Expert: Tran Minh Tuyen	2
3	Stakeholder Engagement Expert 1: Luu Minh Nghia	10
4	International Capacity and Training Expert 2: Michael Mehling	2
5	National Capacity and Training Expert: Nguyen Tien Hai	2
6	International Carbon Pricing Policy Expert: Victor Escalona Gomez	2
7	Survey Data Collection Expert: Dao Xuan Hoach	3
8	IT Expert: Duong Chi Cong	2
9	Supporting experts (including administration and translation)	

3.4.4 Deliverables

The key deliverables in English and Vietnamese include:

- A detailed stakeholder map.
- A detailed stakeholder engagement plan.

If it is requested, the followings are supposed to be submitted together:

- A public facing, publishable Executive Summary (no more than 2 pages) in professional English must be submitted with each deliverable.

- A public facing, catchy PowerPoint presentation highlighting key information must be submitted with each deliverable.

3.5 Task 5: Preparing and providing access to a carbon simulation tool suitable for use under this Assignment

3.5.1 Objectives

The objective of this task is to create and provide access to a customized, carbon market simulation tool specifically designed for Vietnam, which is user-friendly, well-designed, and effectively administered to produce reliable results.

3.5.2 Activities

3.5.2.1 Activity 5.1 Providing access to and identifying options for adjustment of the carbon simulation tool suitable for use for this Assignment

The Consultant will utilize CarbonSim as the simulation tool for the training and simulation activities within this Assignment. CarbonSim is a well-designed and well-administered tool that effectively demonstrates how an ETS can cap and reduce GHG emissions, leverage market forces to cost-effectively reduce emissions, stimulate investment in innovative technologies, contribute to the achievement of economic and social goals. Developed and owned by EDF, CarbonSim has been successfully employed in various locations worldwide, including Viet Nam and a number of other Asian nations, for the benefit of ETS stakeholders.

CarbonSim is widely recognized as one of the most powerful and effective capacity-building carbon market simulation tools available. Its success is demonstrated by its selection as the preferred simulation tool for the international ETS training courses conducted by the ICAP.

Given its proven track record, CarbonSim offers a ready-to-use solution, eliminating the need for additional time and resources to develop a new simulation tool. The Consultant will make necessary adjustments to tailor CarbonSim specifically for this project.

The main reasons for selecting CarbonSim as the simulation tool of choice are as follows:

- The development of a new purpose-built carbon simulation tool would incur a substantial additional cost, estimated to be at least US\$500,000.
- Creating and testing a new tool from scratch could require 12 months.

Considering these factors, the use of CarbonSim offers a proven and successful simulation tool and provides a cost-effective and time-efficient solution. By leveraging CarbonSim for this project, the Consultant can save substantial costs and avoid potential delays associated with developing a new carbon simulation tool.

The CarbonSim tool offers several key features that make it an ideal choice for simulating a real-time compliance market within a government-administered ETS. These features align well with the objectives of this Assignment. The core features of CarbonSim include:

- Can be either administered on-site (via the web or a LAN where web access is not possible) or remotely administered and /or accessed;
- Can accommodate between 36 (with one server and one person teams), 72 (with one server and two person teams...the recommended configuration for this project) or ~300+ users

(with three person teams and three servers running three different simulations simultaneously);

- Simulates select components of emissions trading, which can be introduced in a phased approach:
 - ❖ adjustable and automatic allocation of emissions allowances.
 - ❖ surrender and banking of emissions allowances and offsets (the system will be set to surrender allowances and offsets automatically).
 - ❖ compliance with an adjustable progressive emissions cap.
 - ❖ participation in primary market allowance auctions with adjustable price floors, ceilings, and durations.
 - ❖ use and trading of offsets with adjustable limits.
 - ❖ undertaking abatement measures.
 - ❖ ability to participate in a secondary market (both using an exchange with adjustable exchange volatility limits and an over-the-counter market) for spot transactions.
 - ❖ application of adjustable penalties for non-compliance.
- Features built-in Artificial Intelligence (AI) enhanced capability that allows for simultaneous market participation by firms by both human and AI driven players.
- Allows selection for facilities to be operated either by human players (carbon market exercise participants) or in AI mode.
- Provides for periodic evaluation of facility and system performance.
- Allows for simultaneous operation in different languages and multiple currencies.

Moreover, the tool encompasses an array of additional features that enhance the tool's value proposition, elevating its functionality, realism, maturity, and advancement beyond other simulation tools potentially available to the client. These supplementary features significantly augment the tool's capabilities, ensuring it is more comprehensive, practical, sophisticated, and cutting-edge than alternative options and include:

- Adjustable duration of virtual compliance periods and other time-based components according to the needs of the simulation.
- Adjustable auctions parameters, including the number of auctions, duration of auctions, type and amount of product to be auctioned, price collar, and the interval between auctions, according to the needs of the simulation.
- The ability for both humans and AI players to participate in the auctions and the exchange.
- The use of distinct, sequential simulation periods with changing parameters.
- Accommodates the introduction of changes in the simulation parameters, otherwise known as "shocks" (e.g., changes in allocations, emission growth rates, offset use, etc.) within a simulation period.
- Provides a continuous feed of real-time market information to simulation participants.
- Provides administrators to access user screens, the ability to easily troubleshoot most problems, and effectively demonstrate teaching points.

Table 5: Summary of CarbonSim key specifications

	Function	Comment
Core functions	Abatement	Players can mitigate emissions liability using on-site abatements, which include info on total reductions, cost, and efficacy
	Artificial Intelligence	Affords participation by both AI and human players
	Currency	Multiple currencies can be used simultaneously
	Company management page	Allows players to manage multiple units of the same company
	Dashboard	Provides players with one screen to monitor all critical parameters
	Glossary	Players have access to a hover over key term definitions
	Language	The tool can be simultaneously played in Vietnamese (after the creation of a Vietnamese language version)
	Abatement	Ranks player performance based on marginal cost of control and compliance status in the current year and overall years
	Messaging	Participants can electronically message each other
	Primary market	Allowances can be made available through auctions
	Rules screen	The system allows players to see ETS configurations and rules
	Secondary market	Exchange and the over-the-counter market are provided through which participants can transact allowances and offsets
	System-wide reports	Provides performance metrics that describe the current year and overall metrics regarding emissions, market activity, monies raised, offsets used, penalties assessed, etc.
Administrative Functions	Abatements available	Provides an inventory of abatements with information regarding their name, cost, time to implement, reductions gained, etc.
	Artificial Intelligence trade assistance	Participants can set their companies on AI-mode which will automatically manage their carbon portfolios.
	Allocations	Allows adjustment of the allocations that are provided without charge and the consequential shortfall.
	Allowance surrender	Allowance surrender can be done automatically (the default mode) or manually.

Function		Comment
	Auction	The number, frequency, timing, amount/type of allowances auctioned can be set.
	Banking	The amount of banking (relative to the compliance obligation) can be set.
	Duration of virtual years and simulation	The actual elapsed time of each virtual year can be set from minutes to months.
	Exchange market access	Participants can access a multi-bid and offer exchange market which accommodates the use of market, limit, and stop loss orders.
	Exchange price variation limit	A limit can be set that minimizes exchange volatility.
	Fines	Financial penalties can be assessed against participants that engage in illegal behavior.
	Government reserve	The amount of allowances withheld for auction can be adjusted.
	Leaderboard	The leaderboard can be made visible or hidden from players.
	Monitoring	The administrator can remotely access, monitor, and control the status and actions of players.
	Multiple systems	The administrator can run multiple, simultaneous, and linked simulations.
	Number of years in a simulation	Simulations can be run from one to six or more virtual years.
	Number of players	From one to more than 200 players can be accommodated, depending on the number of players per team (two players per team are recommended) and the number of servers that are utilized.
	Offsets	Both the amount of offsets available and the relative percentage useful to satisfy compliance obligations can be adjusted.
	OTC market access	Access to the over-the-counter market can be turned on or off.
	Penalty multiplier	Excess emissions can be penalized with adjustable monetary fines and next year allowance deductions.
	Player's ability to borrow money	Players can borrow money at an adjustable interest rate.
	Player emission growth	Business as usual emission growth is set typically at rates between two and six percent per year.

Function		Comment
	Price collar	An auction floor and ceiling price (price collar) is set.
	Rate of reduction	The system-wide emission reduction rate can be set at between 0 to >10%/year.
	Registration	The administrator can monitor the registration status of players.
	Time bar and stop trade notification	Players are provided with a time bar that indicates how much time is left in the auction(s) and/or years and notifications indicating that trading has been halted.

3.5.2.2 Activity 5.2: Adjusting the CarbonSim tool

As mentioned earlier, the primary purpose of the CarbonSim tool is to demonstrate the principles of emissions trading and bring markets to life. ETS program administrators benefit because it demonstrates that program results are driven by design choices made by policymakers. Industries benefit because it demystifies how to develop and implement a carbon portfolio management strategy. It is important to note that any quantitative results should not be considered for policymaking purposes. To tailor the tool to the national context, the Consultant will implement the following adjustments:

- Translating into Vietnamese: The tool will be translated into the Vietnamese language to ensure a common understanding among participants.
- Adjusting the currency to Vietnamese Dong: This adjustment aims to expedite the decision-making process during the simulation by using the local currency. Note, while players will be free to use either Dong or USD, given the Dong to USD exchange rate (currently ~23,825 : 1), we anticipate that the administrator will run the exercises in USD.
- Other parameters: The Consultant will make adjustments to certain other parameters based on participants' feedback after each simulation, ensuring that the tool is continuously refined and aligned with their needs.

These adjustments aim to enhance the tool's effectiveness and relevance within the specific context of the Assignment, providing participants with a more tailored and meaningful simulation experience.

3.5.2.3 Activity 5.3: Testing the tool

Following the incorporation of each adjustment round into the simulation, a beta test will be conducted with the participation of the Consultant's experts as participants to ensure the functionality and stability of the tool. It is important to note that adjustments leading to system instability and/or which require an extraordinary amount of time/resources to implement cannot be accommodated.

In this regard, activities 5.2.2 and 5.2.3 may be repeated as necessary after each simulation training to ensure the tool's stability following any adjustments. It should be clarified that certain 'on-the-fly' mid-simulation exercise adjustments can be made without the need for beta testing. The goal

is to maintain the tool's stability while continuously refining its functionality based on feedback and testing.

3.5.3 Allocation of human resources

Details of human resources of this Task are provided below:

Table 6: Details of human resources for Task 5 (W5- W23, 01 Sep -21 Jan 2024)

No	Personnel and total time inputs	Person days
1	Team Leader/Climate Change Policy Expert: Dang Hong Hanh	8
2	National GHG Inventory/Mitigation Expert: Tran Minh Tuyen	10
3	Carbon Market Simulation Expert: Josh Margolis	10
4	National Capacity and Training Expert: Nguyen Tien Hai	10
5	Survey Data Collection Expert: Dao Xuan Hoach	10
6	IT Expert: Duong Chi Cong	10
7	Supporting experts (including administration and translation)	

3.5.4 Deliverables

The key deliverable includes access to ready-to-use carbon market simulation tool for the training sessions.

If it is requested, the following will be submitted together:

- A public facing, publishable Executive Summary (no more than 2 pages) in professional English must be submitted with each deliverable.
- A public facing, catchy PowerPoint presentation highlighting key information must be submitted with each deliverable.

3.6 Task 6: Providing a web-based carbon simulation platform

3.6.1 Objectives

The Objectives of this Task is to develop a web-based platform serves multiple purposes within the Assignment. It acts as an update hub for the training sessions and activities, providing essential information and resources related to the ETS. Additionally, the platform serves as a knowledge hub, enhancing the readiness of stakeholders to participate in the simulations. It also acts as a channel to increase public awareness regarding ETS and carbon pricing instruments. Overall, the web-based platform plays a crucial role in strengthening stakeholder readiness, facilitating information exchange, and promoting public awareness on ETS and carbon pricing instruments.

3.6.2 Activities

3.6.2.1 Activity 6.1: Developing the web-based platform

The Consultant will develop (and/or enhance) the web-based platform comprising three sub-components as outlined in the figure below:

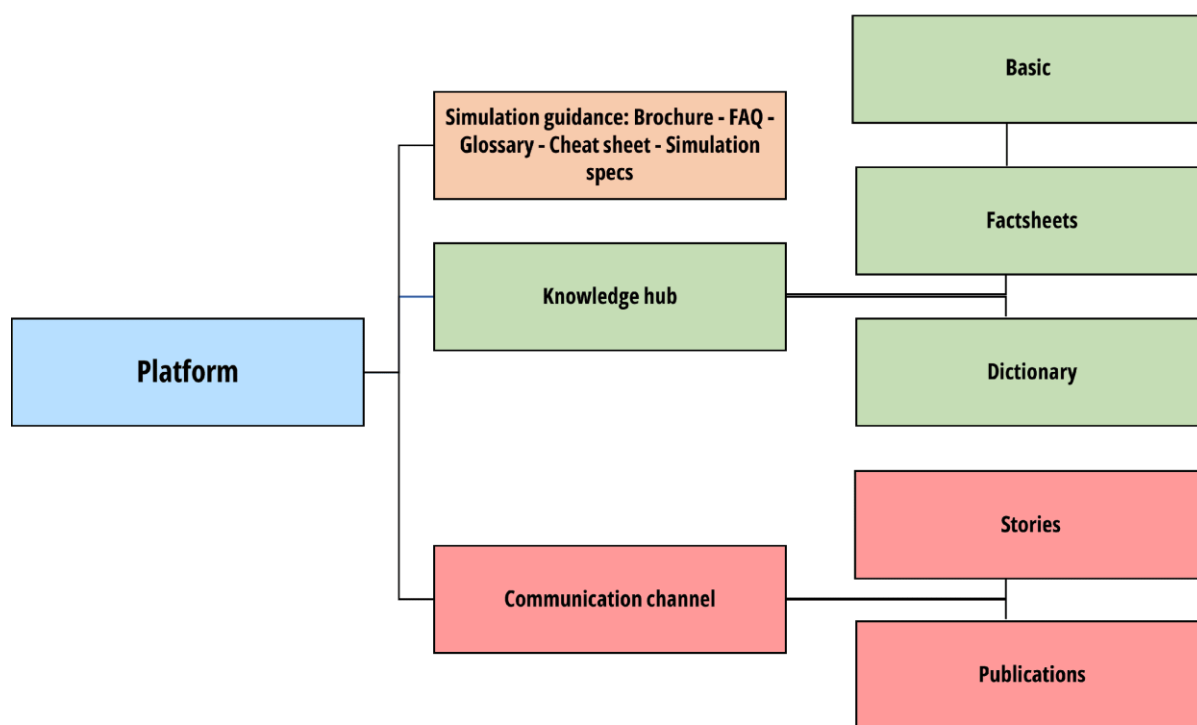


Figure 7: Structure of the web-platform

The web-based platform will consist of the following three sub-components:

- **Simulation Guidance:** This component will provide participants with comprehensive technical guidance in advance through a Vietnamese language brochure, FAQ section, glossary, cheat sheet, and simulation specifications. These resources will help participants familiarize themselves with the tool and its functionalities.
- **Knowledge Hub:** The knowledge hub will serve as a repository of systematic information related to climate change, with a specific focus on the carbon market and ETS. It will include the following components:
 - ❖ **Basic:** This section will cover the fundamentals of ETS and carbon-pricing systems.
 - ❖ **Factsheets:** Participants can delve into specific carbon-related topics across both voluntary and compliance markets.
 - ❖ **Dictionary:** This feature will provide explanations of commonly used words, phrases, and acronyms related to carbon pricing and markets.
- **Communication Channel:** This component will function as a forum for disseminating updates and information about the Assignment, its activities, outcomes, and developments in the Viet Nam and international carbon markets. It will include the following elements:
 - ❖ **Stories:** This section will provide news updates about the state and trends of the international and Viet Nam carbon market.
 - ❖ **Publications:** Participants can access updates on the activities and technical reports of the Assignment.

3.6.2.2 Activity 6.2: Developing the content for the web-based platform

The Consultant will closely consult with ETP and DCC to develop the content for the web-based platform, incorporating the most up-to-date information on the global development and operation of ETS, as well as domestic experiences.

These contents of the sub-components will collectively address the awareness and readiness gaps identified in Task 3 and significantly enhance participants' understanding, knowledge, and engagement in the simulations. By aligning the content of the knowledge hub with these identified gaps, the platform will effectively fulfill the needs of the participants, ensuring that they have the necessary information and resources to actively participate in the simulations. Additionally, the communication channel will serve as a reliable source for relevant updates and information, keeping participants informed about the progress and developments of the Assignment. The platform's content development and operation will be closely consulted with ETP and DCC.

3.6.2.3 Activity 6.3: Updating the web-based platform

Throughout the project timeline, the platform will undergo regular updates in accordance with the Project's Activities. Once the project is completed, the platform can continue to serve as a valuable knowledge hub. In line with this, the Consultant will hand over the platform to DCC, who can integrate it into their website following the launch of the ETS pilot. This transition ensures the sustained availability of the platform's resources and information for ongoing use and dissemination within the context of the ETS implementation.

3.6.3 Allocation of human resources

Details of human resources of this Task are provided below:

Table 7: Details of human resources for Task 6 (W9- W24, 01 Oct -31 Jan 2024)

No	Personnel and total time inputs	Person days
1	Team Leader/Climate Change Policy Expert: Dang Hong Hanh	8
2	National GHG Inventory/Mitigation Expert: Tran Minh Tuyen	8
3	Carbon Market Simulation Expert: Josh Margolis	3
4	International Capacity and Training Expert 2: Michael Mehling	4
5	National Capacity and Training Expert: Nguyen Tien Hai	8
6	International Carbon Pricing Policy Expert: Victor Escalona Gomez	4
7	Survey Data Collection Expert: Dao Xuan Hoach	15
8	IT Expert: Duong Chi Cong	15
9	Supporting experts (including administration and translation)	

3.6.4 Deliverables (Milestone 2)

- A detailed stakeholder map.
- An Engagement Plan satisfactory to and agreed upon by the DCC and the ETP.
- A readiness assessment report.
- A list of potential players in the future ETS that will participate in the trainings.
- A web-based platform that can be accessed freely before the trainings and maintaining/updating regularly throughout the implementation of this Assignment.
- A ready-to-use CarbonSim tool for the simulation that is tailored made for Vietnam.

The following will be submitted together with the above deliverables:

- A public facing, publishable Executive Summary (no more than 2 pages) in professional English must be submitted with each deliverable.
- A public facing, catchy PowerPoint presentation highlighting key information must be submitted with each deliverable.

3.7 Task 7: Organizing a series of ETS simulation training/workshops/forums

3.7.1 Objectives

The objective of this activity is to enhance ETS literacy, strengthen capacity, and accelerate the implementation of an ETS for key government officers and emitters. This will be achieved through a comprehensive program of trainings and simulations designed to facilitate learning and understanding of ETS principles and processes.

3.7.2 Activities

3.7.2.1 Activity 7.1: Pre-training activities

The preparation phase encompasses the following activities:

- Developing the agenda based on the outcomes of Task 3 and Task 4.
- Onboarding/registering the participants: Collaborating closely with the potential players identified in Task 3 to confirm their participation in the training sessions.
- Establishing the baseline parameters for the simulations.
- Conducting training sessions for the trainers (also known as tutors) to provide them with the necessary skills and knowledge to support and facilitate the training sessions and simulations.
- Managing all logistical aspects of the trainings, including identifying participants who will receive financial support to attend (priority criteria will be proposed in Task 3, taking into account factors such as gender balance, geographic representation, size and sectoral diversity, etc.).

3.7.2.2 Activity 7.2: Organizing the training sessions

The proposed training program consists of four courses:

- **The first course**, taking place in Ha Noi, will span two days of intensive training. It will primarily target 50 key government officers and 20 stakeholders from the North, including NPOs, NGOs, financial institutions, banks, and research organizations. Given the fact that the central government officers who are responsible for the ETS design and management are located in Ha Noi, this course will prioritize design adjustments to highlight the impact of design decisions on the ETS's performance and the regulated emitters.
- **The second course**, also held in Ha Noi, will be a two-day intensive training program focused on regulated emitters from the North and Central Viet Nam. The expected number of participants for this course is 100.
- **The third and fourth courses**, each lasting two days, will be conducted in Ho Chi Minh City. These trainings will specifically cater to ~140 regulated emitters in the South, along with some participants from Central Viet Nam (Figure 8).

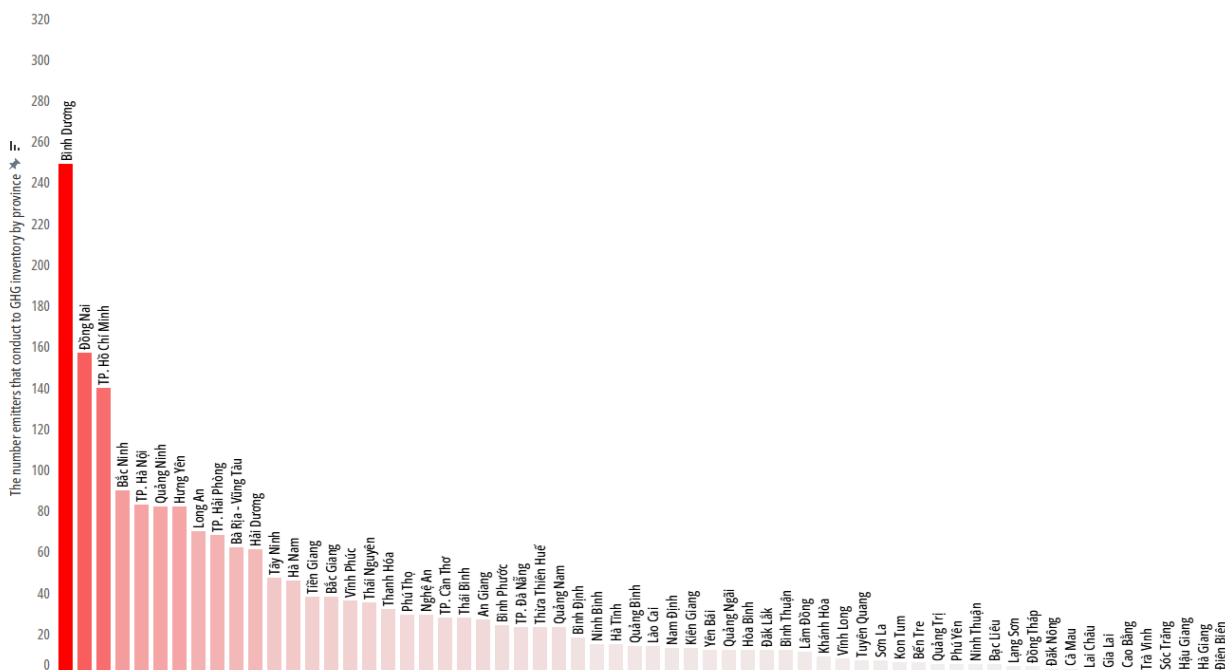


Figure 8: The number of GHG emitters that conduct to GHG inventory based on Decision 01/2022/QĐ-TTg Dated 18 Jan. 2022

The first and second courses will be scheduled to be consecutively organized in the second week of April, and the same arrangement will be applied to the last two courses which are scheduled for the second week of June. This allows for efficient use of time and resources.

Additionally, there is a possibility of organizing an additional simulation specifically for university students at no cost, contingent upon the timeline and agreement between ETP and DCC.

The rationale for structuring the courses as described above includes:

- The first course will prioritize key government officers to gather their insights, desires, and needs, which will serve as valuable inputs for Task 8. Taking into account the timeline of Task 8.
- By considering the geographic distribution of regulated emitters, including international travel for experts and domestic travel for the team and participants, the aim is to minimize travel costs incurred.
- The structure ensures equal opportunities for participation in the training sessions, providing access to potential players throughout the entire country.

The summary of the structure of this training is shown in the figure below:

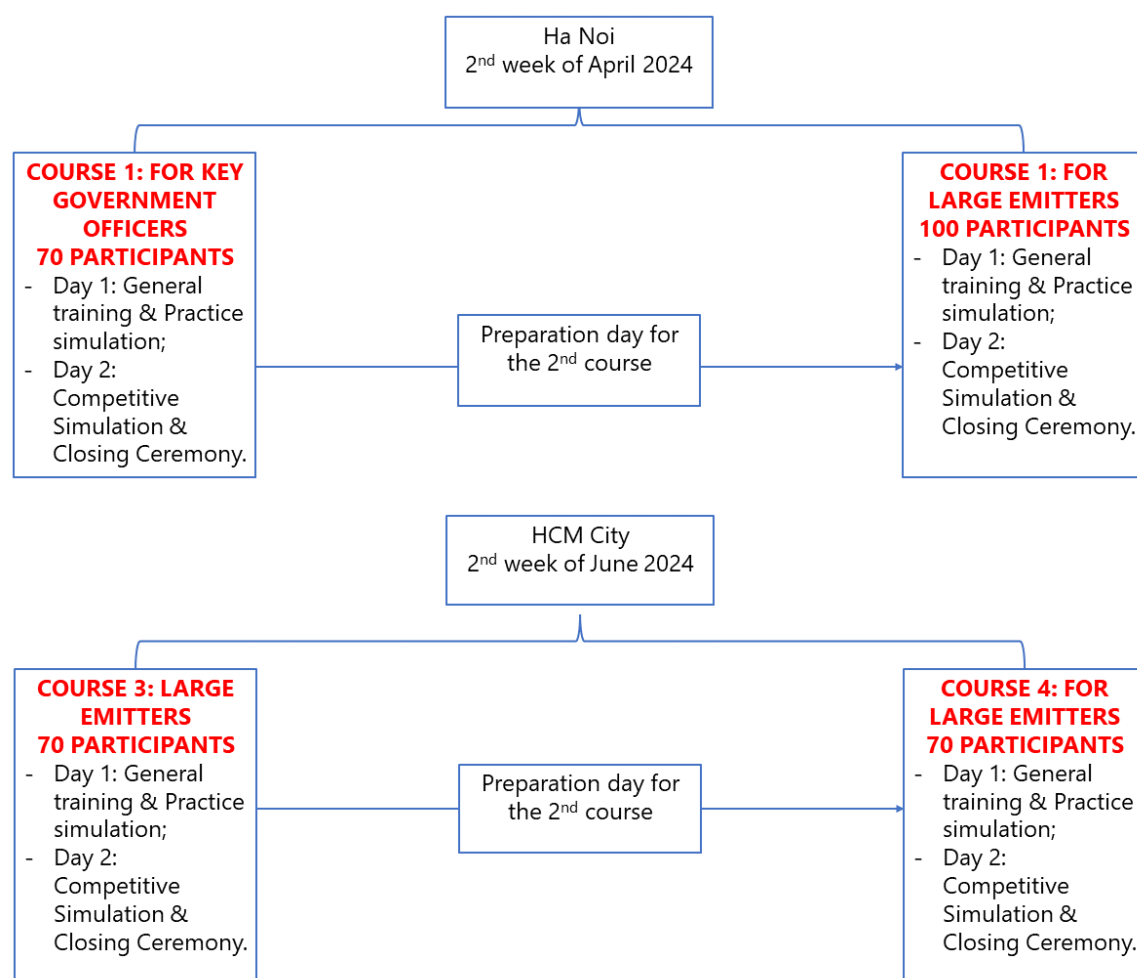


Figure 9: Structure of the trainings

During the Inception Workshop, it is requested from both DCC and ETP to the Consultant to facilitate the preparation in order to organize the training courses earlier than the initial plan. Therefore, the Consultant will aim to organize the first courses in Hanoi in February/March and then the courses in HCM city in March/April. The specific dates will be consulted early with both DCC and ETP.

Each training session will consist of two sequential half day sections:

- A general section, dedicated to introducing the theory of ETS delivered by ETS experts; and
- An ETS simulation practice utilizing CarbonSim delivered by CarbonSim admin.

The table below displays the Flow Diagram for the typical simulation exercise in each training section:

Table 8: Typical Simulation Exercise Flow Diagram

Period	Consultant	Participants
Day 1	Commence Day 1 PRACTICE simulation to familiarise participants with each other and the workings of (and screens associated with) CarbonSim, as well as to provide them with an introduction to basic ETS terminology (e.g., cap, trade, allowances, offset, banking, auction, exchange, OTC, etc.).	

	Provide participants with lecture as to: ● Background as to value/limits of simulations; ● ETS principles; ● CarbonSim principles; ● Recommendations as to effective strategies; ● Provide registration instructions; ● Open Sim for registrations; ● Troubleshoot/resolve registration problems/remove duplications; ● Divided participants into smaller groups of 2-3 people.	Respond to instructor prompts as appropriate: ● Complete the entrance survey/test; ● Register.			
	Start Year #1				
	● Monitor system; ● Monitor individuals; ● Detect/resolve problems; ● Render assistance; ● Provide liquidity as necessary.	Abatement	Four auction rounds	Exchange	OTC
	Year #1 ends				
	End year #1 ● Calculate system-wide totals; ● Calculate individual performance; ● Highlight teaching points; ● Prepare for Year #2; ● Consider results in light of teaching objectives and participant performance; ● Decide on appropriate adjustments/shocks; ● Beta-test shocks (time allowing); ● Implement shocks.	● Take in the information provided by Sim-administrator; ● Respond to discussion prompts; ● Share strategies; ● Ask questions; ● Reflect on Year #1; ● Consider lessons learned; ● Initiate internal dialogues reconnection between simulation experiences and actual ETS stakeholder discussions.			
	Commence Year #n				
	● Monitor system; ● Monitor individuals; ● Detect/resolve problems;	● Take in the information provided by Sim-administrator;			

	● Render assistance; ● Provide liquidity as necessary.	● Respond to discussion prompts; ● Share strategies; ● Ask questions; ● Reflect on Year #1 – n; ● Consider lessons learned; ● Initiate internal dialogues regarding the connection between simulation experiences and actual ETS stakeholder discussions; ● Complete the exit survey/test.			
	Day 1 Practice Simulation				
	For the entire simulation: ● Calculate system-wide totals; ● Calculate individual performance; ● Highlight teaching points; ● Review key terms; ● Extrapolate Sim results to real life; ● Review the schedule for day 2.	Participate in discussion and respond to administrator prompts.			
Day 2	Commence Day 2 COMPETITIVE Simulation to provide the opportunity to apply the lessons they learned from the first exercise and advised that the top performers – those who compiled at the lowest marginal cost of control – at the end of the simulation would be recognized in the award ceremony.				
	● Provide registration instructions; ● Open Sim for registrations; ● Troubleshoot/resolve registration; ● Complete the entrance survey; ● Register.				
	Start Year #1				
	● Monitor system; ● Monitor individuals; ● Detect/resolve problems; ● Render assistance; ● Provide liquidity as necessary.	Abatem ents	Four auction rounds	Exchan ge	OTC
Year #1 ends					

	End Year #1	● Take the information provided by Sim-administrator; ● Respond to discussion prompts; ● Share strategies; ● Ask questions; ● Reflect on Year #1; ● Consider lessons learned; ● Initiate internal dialogues regarding the connection between simulation experiences and actual ETS stakeholder discussions.			
	Commence Year #n				
	● Monitor system; ● Monitor individuals; ● Detect/resolve problems; ● Render assistance; ● Provide liquidity as necessary.	Abatements	Four auction rounds	Exchange	OTC
	Year #n ends				
	End Year #n: ● Calculate system-wide totals; ● Calculate individual performance; ● Highlight teaching points. Prepare for Year #n: ● Consider results in light of teaching objectives and participant performance; ● Decide on appropriate adjustments/shocks; ● Beta-test shocks (time allowing); ● Implement shocks.	● Take in the information provided by Sim-administrator; ● Respond to discussion prompts; ● Share strategies; ● Ask questions; ● Reflect on Year #1 – n; ● Consider lessons learned; ● Initiate internal dialogues regarding the connection between simulation experiences and actual ETS stakeholder discussions; ● Complete the exit survey/test.			
	Day 2 competitive simulation ends				
	For the entire simulation: ● Calculate system-wide totals;	Participate in discussion and respond to administrator prompts			

	Calculate individual performance.	
	Prepare and deliver closing and award ceremony	
	With participants: - Highlight teaching points; - Review key terms; - Extrapolate Sim results to real life; - Recognize top performers; - Put simulation experience into the context of jurisdictional ETS development activities; - Emphasize results are a function of effective design and administration and capacity building; - Administer exit survey/test for participants that have yet to complete.	Complete the exit survey/test (for those that have yet to complete it).

Following the completion of each training course, participants will complete a survey or evaluation form to assess their knowledge and understanding. These results will then be compared with the awareness and readiness outcomes identified in Task 3 and the pre-course survey, enabling the monitoring of training effectiveness and the identification of valuable lessons learned.

Training sessions will also be recorded to be kept as training materials and proceedings and available for later use.

3.7.3 Allocation of human resources

Details of human resources of this Task are provided below:

Table 9: Details of human resources for Task 7 (W25- W48, 01 Feb -31 Jul 2024)

No	Personnel and total time inputs	Person days
1	Team Leader/Climate Change Policy Expert: Dang Hong Hanh	10
2	National GHG Inventory/Mitigation Expert: Tran Minh Tuyen	10
3	International GHG Inventory/Mitigation Expert: Roxanne Tan	8
4	Carbon Market Simulation Expert: Josh Margolis	16
5	Stakeholder Engagement Expert 1: Luu Minh Nghia	10
6	International Capacity and Training Expert 1: Karolien Casaer-Diez	8
7	International Capacity and Training Expert 2: Michael Mehling	7
8	National Capacity and Training Expert: Nguyen Tien Hai	10
9	International Carbon Pricing Policy Expert: Victor Escalona Gomez	8
10	Survey Data Collection Expert: Dao Xuan Hoach	14
11	IT Expert: Duong Chi Cong	14
12	Supporting experts (including administration and translation)	

3.7.4 Deliverables

- A training report of each set of courses (one report for the simulations run in Hanoi and a second report for the simulations run in Ho Chi Minh City).
- Training materials and post-course assessment regarding awareness and readiness of the participants toward the ETS.

If it is requested, the followings are supposed to be submitted together:

- A public facing, publishable Executive Summary (no more than 2 pages) in professional English must be submitted with each deliverable.
- A public facing, catchy PowerPoint presentation highlighting key information must be submitted with each deliverable.

3.8 Task 8: Planning a carbon market master class for key government officials to learn and exchange governance knowledge and experience of ETS.

3.8.1 Objectives

The aim of this activity is to assess the requirements and collaborate with the ETP and DCC in order to develop an on-the-job training course tailored for key government officials from MONRE and other relevant ministries (such as the Ministry of Finance, Ministry of Planning and Investment, and Ministry of Industry and Trade). This training course will be conducted in one or two countries where an emission trading exchange already exists, or in a country that shares conditions similar to those in Viet Nam, but that has advanced experience in ETS implementation.

3.8.2 Activities

3.8.2.1 Activity 8.1: Identify capacity training needs/gaps of government officials

The Consultant will consolidate the outputs, feedback, and performance evaluations of the participants from the initial training course. Based on this analysis, a comprehensive list of topics will be proposed for further exploration in the master class, with a specific focus on the design and operation of the ETS. Additionally, topics that cannot be readily covered in a training course and are more effectively addressed in direct exchange with decision-maker counterparts will be included in the proposed list of topics. A tentative list of participants or agencies will be identified to ensure the targeted inclusion of relevant stakeholders.

3.8.2.2 Activity 8.2: Design the study tour/ master class in close consultation with ETP and DCC

The Consultant will undertake a comprehensive review of the design and operation of ETS systems in the world and select international ETS markets. This analysis aims to identify the strengths, limitations, and parallels between these markets and the intended design of Viet Nam's future ETS. The benchmarking criteria for selection of ETS systems for the master class/ study tour include: 1) the ETS is from ETP's Funders' countries, 2) ETSs in countries with comparable socio-economic context with Viet Nam, 3) ETSs that are functioning well and shows effectiveness, and ETSs that are of preference of Viet Nam's government officials who directly involve in the development of ETS in Viet Nam. Based on this evaluation criteria, one or two countries will be proposed as potential host locations for the master class, considering their relevance and applicability to the context of Viet Nam. The selection of countries will also consider logistical constraints and the

availability of key stakeholders in the host country to contribute to effective implementation of the study tour and master class.



Figure 10: Workflow under Task 8

3.8.3 Allocation of human resources

Details of human resources of this Task are provided below:

Table 10: Details of human resources for Task 8 (W5- W12, 01 Sep -31 Oct 2023)

No	Personnel and total time inputs	Person days
1	Team Leader/Climate Change Policy Expert: Dang Hong Hanh	7
2	National GHG Inventory/Mitigation Expert: Tran Minh Tuyen	1
3	International GHG Inventory/Mitigation Expert: Roxanne Tan	8
4	Stakeholder Engagement Expert 1: Luu Minh Nghia	4
5	International Capacity and Training Expert 1: Karolien Casaer-Diez	8
6	International Capacity and Training Expert 2: Michael Mehling	3
7	National Capacity and Training Expert: Nguyen Tien Hai	1
8	Supporting experts (including administration and translation)	

3.8.4 Deliverables

A plan/proposal of a study tour on ETS design and implementation.

If requested, the following deliverables are supposed to be submitted together:

- A public facing, publishable Executive Summary (no more than 2 pages) in professional English must be submitted with each deliverable.
- A public facing, catchy PowerPoint presentation highlighting key information must be submitted with each deliverable.

3.9 Task 9: Conducting a technical report summarising the results of the simulation and impact assessments

3.9.1 Objectives

The goal of this task is to consolidate the outcomes derived from the simulations and trainings in order to generate comprehensive recommendations and valuable insights for the future implementation of Viet Nam's ETS.

3.9.2 Activities

3.9.2.1 Activity 9.1: Composing the technical report

Upon completion of the second set of courses, an in-depth report will be compiled to provide a comprehensive summary of the lessons learned, progress made, and recommendations derived throughout the duration of this Assignment. The technical report will encompass various sections, including but not limited to:

Box 2: Preliminary outline of the technical report

- Introduction; - Objectives and purpose of the trainings; - Demographics of the participants; - Overview of the training methodology; - Assessment of participants' ETS knowledge acquired through the simulations; - Description of the simulation tools utilized; - Analysis and presentation of simulation results; - Evaluation of project outcomes and necessary adjustments; - Key lessons learned and recommendations for the future implementation of Viet Nam's ETS; - Proposed next steps to advance the initiative.
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3.9.2.2 Activity 9.2: Organizing the High-level consultation

The draft technical report will undergo a high-level consultation process, involving selected participants with valuable insights and expertise. This consultation will include:

- Trainees who demonstrated outstanding performance in previous courses;
- Key government officials responsible for the design and management of the future ETS;
- Research organizations, think tanks, NPOs, NGOs, and other relevant stakeholders.

This consultation will facilitate an in-depth discussion aimed at finalizing the Technical Report, ensuring its accuracy and relevance. The input and feedback gathered from these participants will be instrumental in refining and enhancing the report's content.

3.9.3 Allocation of human resources

Details of human resources of this Task are provided below:

Table 11: Details of human resources for Task 9 (W37- W40, W45 – W64, 01 May -31 May, 01 Jul – 30 Nov, 2024))

No	Personnel and total time inputs	Person days
1	Team Leader/Climate Change Policy Expert: Dang Hong Hanh	7
2	International GHG Inventory/Mitigation Expert: Roxanne Tan	2
3	Carbon Market Simulation Expert: Josh Margolis	11
4	Stakeholder Engagement Expert 1: Luu Minh Nghia	3
5	International Capacity and Training Expert 1: Karolien Casaer-Diez	2
6	International Capacity and Training Expert 2: Michael Mehling	7
7	National Capacity and Training Expert: Nguyen Tien Hai	8
8	International Carbon Pricing Policy Expert: Victor Escalona Gomez	2

9	Survey Data Collection Expert: Dao Xuan Hoach	3
10	IT Expert: Duong Chi Cong	3
11	Supporting experts (including administration and translation)	

3.9.4 Deliverables (Milestone 3)

Key deliverables in English and Vietnamese include:

- A training report of each set of courses (one report for the simulations run in Hanoi and a second report for the simulations run in Ho Chi Minh City).
- Training materials and post-course assessment regarding awareness and readiness of the participants toward the ETS.
- Draft plan/proposal of a study tour on ETS design and implementation.
- Technical report summarizing the results of the simulation and impact assessments.

Together with the deliverables, the followings are supposed to be submitted by the Consultant:

- A public facing, publishable Executive Summary (no more than 2 pages) in professional English must be submitted with each deliverable.
- A public facing, catchy PowerPoint presentation highlighting key information must be submitted with each deliverable.

3.10 Task 10: Final report and final workshops

3.10.1 Objectives

The purpose of this task is to consolidate and analyze the outcomes of the simulations, with a specific focus on identifying key findings and generating actionable recommendations for the future implementation of an ETS in Vietnam. The aim is to extract valuable insights from the simulation exercises and distill them into a comprehensive summary report that can inform and guide the development and design of the country's future ETS. By examining the simulation results in conjunction with relevant factors and considerations, this task will contribute to shaping the roadmap and strategy for a successful and effective ETS in Vietnam.

3.10.2 Activities

3.10.2.1 Activity 10.1: Composing the final closing report

The comprehensive final report serves as a consolidated overview of the entire project, encompassing a cohesive narrative of the deliverables. It is meticulously crafted to adhere to professional standards and can be considered a publishable document with the potential for wide dissemination. Notably, the report incorporates a finalized version of the Results Based Monitoring Framework, which includes a gender-disaggregated analysis to ensure inclusivity and address gender-related considerations.

3.10.2.2 Activity 10.2: Organizing the final workshop

A concluding workshop will be organized to showcase the accomplishments, significant findings, and valuable lessons derived from the implementation of the Assignment. This workshop will serve as a platform to present the outcomes and share insights with relevant stakeholders. Notably, international experts from the Consultant will actively participate in the workshop, offering their expertise and contributing to the knowledge exchange. By facilitating on-site engagement, the

workshop aims to foster meaningful discussions, promote collaboration, and provide an opportunity for participants to gain a deeper understanding of the achievements and lessons learned throughout the Assignment.

3.10.3 Allocation of human resources

Details of human resources of this Task are provided below:

Table 12: Details of human resources for Task 10 (W65- W70, 01 Dec 2024 -10 Jan, 2025)

No	Personnel and total time- inputs	Person days
1	Team Leader/Climate Change Policy Expert: Dang Hong Hanh	9
2	National GHG Inventory/Mitigation Expert: Tran Minh Tuyen	2
3	International GHG Inventory/Mitigation Expert: Roxanne Tan	6
4	Carbon Market Simulation Expert: Josh Margolis	5
5	Stakeholder Engagement Expert 1: Luu Minh Nghia	6
6	International Capacity and Training Expert 1: Karolien Casaer-Diez	6
7	International Capacity and Training Expert 2: Michael Mehling	4
8	National Capacity and Training Expert: Nguyen Tien Hai	2
9	International Carbon Pricing Policy Expert: Victor Escalona Gomez	6
10	Survey Data Collection Expert: Dao Xuan Hoach	3
11	IT Expert: Duong Chi Cong	3
12	Supporting experts (including administration and translation)	

3.10.4 Deliverables (Milestone 4)

Key deliverables in English and Vietnamese include:

- Grand final report.
- Proceedings of the final workshop.

Together with the deliverables, the followings are supposed to be submitted by the Consultant:

- A public facing, publishable Executive Summary (no more than 2 pages) in professional English must be submitted with each deliverable.
- A public facing, catchy PowerPoint presentation highlighting key information must be submitted with each deliverable.
- The consultant is required to include their Gantt chart and the results and achievements of the project in accordance with the agreed project level Results-Based Monitoring Framework, the latter as per the approved template. All results, where applicable, must be gender disaggregated.

4 Timeframe for implementation

The above tasks are organized and executed in a structured workflow as outlined below:

WORK SCHEDULE

No.	Task and Deliverable	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
		2023					2024												
		Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan
Task 0	Project management																		
0.1	Project management, including QA/QC																		
Task 1	Composing inception report and workplan																		
1.1	Internal kick-off team meeting																		
1.2	Kick-off meeting with ETP																		
1.3	Compose inception report																		
D1.1	Minutes of kick-off meeting																		
D1.2	Draft Inception Report																		
Task 2	Organize the inception workshop																		
2.1	Prepare inception workshop																		
2.2	Organize the Inception workshop																		
D2.0	Inception workshop minutes																		
D2.1	Final inception report																		
Milestone 1 completion																			
Task 3	Conduct analysis, surveys and consultation events on readiness and awareness of enterprises and facilities on GHG emission/mitigation and carbon market																		
3.1	Conduct desk review and consultations on the ETS (non-emitter entities) to identify the gap on awareness and readiness of relevant stakeholders regarding the ETS																		
3.2	Design the survey questionnaire																		
3.3	Determine the scale and coverage of the survey (including GHG accounting)																		
3.4	Conduct the survey																		
3.5	Data collection, compilation and analysis																		
3.6	Compose an assessment report on awareness and readiness																		
D3.1	Survey questionnaire																		
D3.2	Assessment report on readiness and awareness																		
Task 4	Develop a detailed stakeholder mapping and engagement plan to approach and engage the targeted groups																		
4.1	Mapping all relevant stakeholders of the project																		
4.2	Compose a detailed stakeholder engagement plan																		
O4.1	A detailed stakeholder map																		
D4.1	A detailed stakeholder engagement plan																		
Task 5	Develop a carbon market simulation tool tailor-made for Vietnam																		
5.1	Identify needs for adjustment of the carbon market simulation tool																		
5.2	Adjust the carbon market simulation tool for Vietnamese players																		
5.3	Test the tool																		
D5.1	A ready-to-use carbon market simulation tool																		
Task 6	Develop a web-based carbon market simulation platform																		
6.1	Build the web-based carbon market simulation platform																		
6.2	Develop contents for the web-based platform																		
6.3	Update the web-based platform																		
D6.1	A web-based platform that can be accessed freely before the trainings																		
Milestone 2 completion																			

Annex I presents an initial list of the pertinent key stakeholders involved in the Assignment, along with the Communication Plan. Furthermore, the figure below provides a summary of key stakeholders and donor mapping.

6 A DONOR COORDINATION STRATEGY

Consultations with other donors will serve to get input from their activities and develop a plan to coordinate with them throughout the entire Assignment and for relevant tasks.

Details of relevant activities regarding the capacity training on ETS by other donors in Viet Nam that have been implemented and/or on-going are summarized in the table below. This section will serve as a part of the stakeholder engagement plan that will be conducted in Task 4.

Table 13: Summary of donors and relevant activities in Vietnam

N o.	Donor	Activities	Focal/Contact points	Relevance to the Assignment
1	World Bank	Project Viet Nam Partnership for Market Readiness (VNPMR)	Focal point: Department of Climate Change, Ministry of Natural Resources and Environment (MONRE) Contact: Mr. Luong Quang Huy Head of Division of GHG Emission Reduction and Ozone Layer Protection Phone: +84 (0)902 870505 Email: huyq98@gmail.com	The VNPMR objective is to provide capacity building for the development, implementation and publishing of policies, state management instruments for Nationally Appropriate Mitigation Actions (NAMAs), establishing market-based instruments (MBI), piloting credited NAMAs and developing a roadmap for participating in the domestic and international carbon markets. The project comprises four main components: Component 1: Research, propose and develop policies and state management instruments related to carbon markets Component 2: Develop a database system on GHG emissions, establish market-based instruments and a roadmap for participating in the carbon markets in the field of solid waste management Component 3: Develop and pilot credited NAMAs, develop a reporting system on NAMAs and a roadmap for participating in the carbon markets for the steel production industry Component 4: Strengthen capacities on management, monitoring, evaluation and implementation of NAMA and credited NAMA activities. The project finished in 2020. However, under this project, results of “CS 1.5: Development of training materials on NAMA, credited NAMA, MRV protocol, carbon pricing, market-based instruments and carbon market” can be utilized for the capacity building activities under this Assignment. The Assignment could gain valuable insights from the VNPMR project. Specifically, based on training materials on NAMA, credited NAMA, MRV protocol, carbon pricing, market-based instruments, and carbon market from the VNPMR project, the assignment can consult to develop fundamental ETS terminology, participant engagement strategies and effective instructional tools for capacity training programs about ETS.
		Viet Nam Partnership for	Mr. Nguyen Van Minh	VNPMI is the continuation of the project VNPMR. The objective is to support the

		Market Implementation Project (VNPMI)	Head of Division Division of Economic and Climate Change Information Email: nvminh.dmhcc@gmail.com	Government with the development and application of a National Crediting Program and ETS and enhance expertise and capacity for their implementation. Outputs include legal and technical support for ETS in the industry sector, support for pilot voluntary domestic trading schemes in steel, cement production and thermal power sectors and the detailed design for mandatory domestic ETS. VNPMI is planned to be implemented from 2023 to 2027. Until now, the project has not been implemented yet.
2	The Asian Development Bank (ADB)	Article 6 Support Facility (A6SF)	Virender Kumar Duggal Principal Climate Change Specialist, ADB Email: vkduggal@adb.org	The A6SF provides technical assistance to Viet Nam through policy development and capacity-building support. As part of this support, ADB will help Viet Nam to review its policy framework and develop recommendations to develop its domestic carbon market and to operationalize Article 6. Thus, A6SF includes policy development, capacity-building, stakeholder engagement, and technical support related to a broader range of climate cooperation mechanisms, which are beyond just ETS. Meanwhile, this assignment specifically focuses on ETS Simulation tool, as well as assesses the feasibility, impacts, and practicalities of implementing an ETS in Viet Nam.
3	The United Nations Development Program (UNDP)	Assessment of applicability of carbon projects under the requirement of Article 6 of Paris Agreement and Viet Nam NDC contribution	Hoang Thanh Ha Technical Expert on Energy and Climate Change Mitigation Email: hoang.thanh.ha@undp.org	This project aims to evaluate the feasibility of carbon projects in Viet Nam under the current mechanisms that comply with Article 6 of the Paris Agreement and support Viet Nam's NDC goal. Based on the evaluation results, the project will develop a set of criteria and suggested procedures for transitioning and selecting projects that can contribute to Viet Nam's NDC. Accordingly, the scope of the project and the assignment are different. While the project focuses on assessing carbon projects' application, the assignment aims to pilot the ETS simulation tool and provide lessons learnt and policy recommendations in implementing the future ETS.
4	The Government	The MOU on Carbon Credits	Ministry of Natural	The MOU signals Singapore's and Viet Nam's commitment to collaborate on

	nt of Singapore	Collaboration signed with Singapore	Resources and Environment Contact to be determined later	carbon credit pilot projects which would support both countries in meeting their climate targets. To implement the pilot projects, both countries will also develop modalities and procedures to enable the transfer of correspondingly adjusted carbon credits that are generated. Accordingly, while both the MOU on Carbon Credits Collaboration with Singapore and the ETS Simulation pertain to carbon-related activities, they have different scopes. The MOU is more focused on bilateral collaboration and potential carbon credit trading between countries, whereas the assignment focuses on assessing the implications of implementing an ETS within Vietnam's domestic carbon market.
5	The German Agency for International Cooperation (GIZ)	Macroeconomic Reforms/Green Growth Program	Macroeconomic Reforms/ Green Growth Program Tel: (+84) 2437345195 - Ext. 101 Email: office.macro@giz.de	This program aims to strengthen the management of its public and private resources towards inclusive and green growth by enhancing the coherence of Viet Nam's economic policies, etc., green economic policies, green fiscal policies, credit policies, banking, and green public financial management. Thus, both the Macroeconomic Reforms/Green Growth Program and the assignment aim to contribute to sustainable development, but they have different areas of focus and levels of specificity. The Macroeconomic Reforms/Green Growth Program is a broader initiative aimed at integrating sustainability into economic development, while the assignment focuses specifically on developing, and piloting an ETS tool and providing recommendations for the future ETS.
6	The United States Agency for International Development (USAID)	USAID Viet Nam Low Emission Energy Program (V-LEEP)	Dr. Jennifer Leisch Director - Advanced Energy Systems, Deloitte Email: jleisch@deloitte.com	V-LEEP helps the Government of Viet Nam (GVN) establish an effective policy, regulatory, and incentive environment for low-emission growth in the energy sector, while simultaneously attracting public-sector and private-sector investment in renewable energy (RE) development and energy efficiency (EE). V-LEEP promotes the development of critical building blocks to scale up clean energy, such as accessible smart incentives for clean energy and EE investments, enabling a competitive environment for RE

				generation, enhancing renewable power grid integration, and ensuring the locational concentration of clean energy generation facilities. Accordingly, compared to the assignment, V-LEEP has a broader scope, promoting a supportive environment for renewable energy and energy efficiency investments.
		USAID Sustainable Forest Management	DAI Global LLC.	USAID Sustainable Forest Management works with the Government of Viet Nam to reduce carbon emissions associated with deforestation, the degradation of natural forests, and poor plantation management. Accordingly, the Project implements a “Green Prosperity” approach that strengthens local communities’ ability to protect their natural resource base and reduce emissions, while building a strong foundation for sustainable forest management, livelihoods, and equitable economic growth. Within the project framework, a training class on ETS was conducted. However, this training was confined to stakeholders specifically within the forest sector. Meanwhile, the assignment aims to organize a series of ETS simulations, encompassing a broader spectrum of participants from governmental authorities and significant emitters from that are regulated under Decision No. 01/2022/QD-TTg, dated January 18, 2022.
7	Japan International Cooperation Agency (JICA)	Support the Planning and Implementation of NAMAs in a MRV Manner (SPI-NAMA)	Focal point: Ministry of Natural Resource and Environment (MONDE) / Department of Climate Change (DCC) Contact: Mr. Luong Quang Huy Head of Division of GHG Emission Reduction and Ozone Layer	The purpose of SPI-NAMA is to enhance the capacity of the Government of Viet Nam concerning the planning and implementing of Nationally Appropriate Mitigation Actions (NAMAs). According to the National Strategy on Climate Change issued by the Prime Minister of Viet Nam in December 2011, Nationally Appropriate Mitigation Actions (NAMAs) will be planned by MONRE. To attain this goal, SPI-NAMA has two project components. The first project component aims to enhance the capacity of MONRE to facilitate the process of development and implementation of NAMAs. The second component aims to enhance the capacity of cities in Viet Nam to quantify greenhouse gas (GHG)

			Protection Phone: +84 (0)902 870505 Email: huylq98@gmail.com	emissions and reductions, taking Ho Chi Minh City (HCMC) as a model city. Thus, the assignment could consult the project's approach in identifying related stakeholders; data sources, and analytical tools used by SPI-NAMA for GHG inventory; SPI-NAMA's experiences, challenges, and successes when implementing the assignment.
		Support for Planning and Implementation of the Nationally Determined Contributions (NDC) in Viet Nam (SPI-NDC)		The project aims to support the Government of Viet Nam in the planning and implementation of the Nationally Determined Contributions (NDC) in Viet Nam, recognized by the development and implementation of GHG mitigation policies and GHG mitigation targets of the private sector. Accordingly, the assignment could consult the GHG emissions inventory tool utilized in SPI-NDC to calculate the average GHG emissions for each sub-sector and provide sub-sectoral thresholds to identify large pollution source emitters to engage in the ETS training programs.

7 PROJECT MANAGEMENT

The figure below depicts the organizational structure and delineation of responsibilities for the implementation of the project:

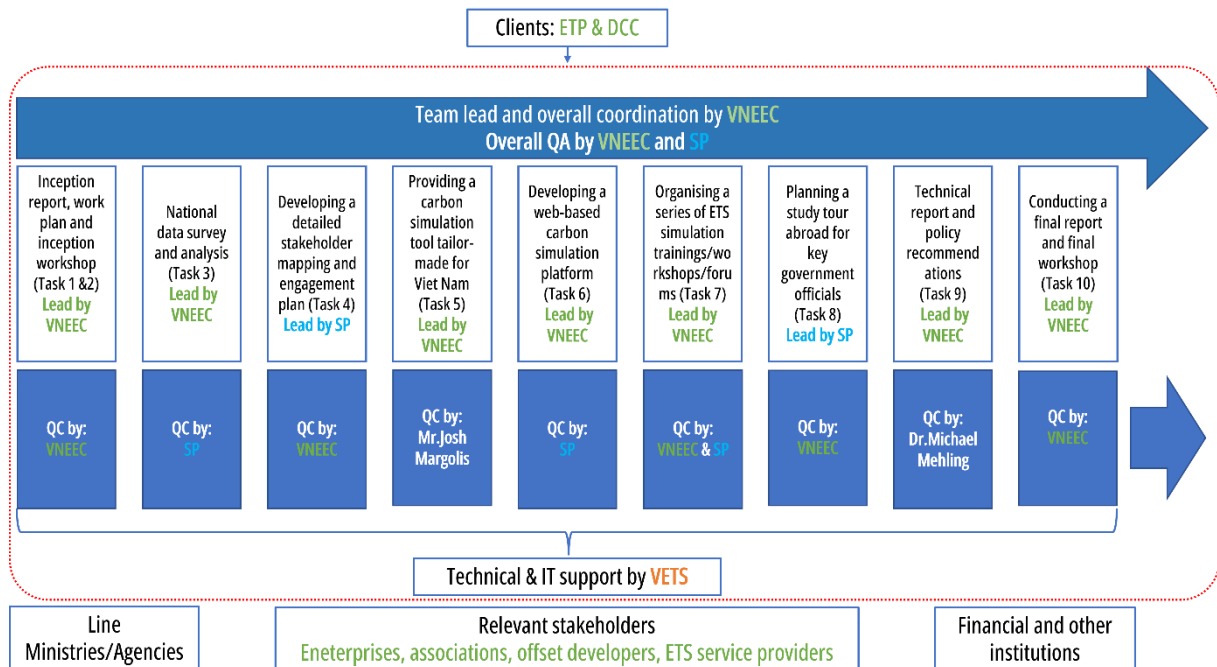


Figure 13: Project implementation and responsibilities by project partners

Detailed information on the key experts of the team and their role are shown in the figure below:

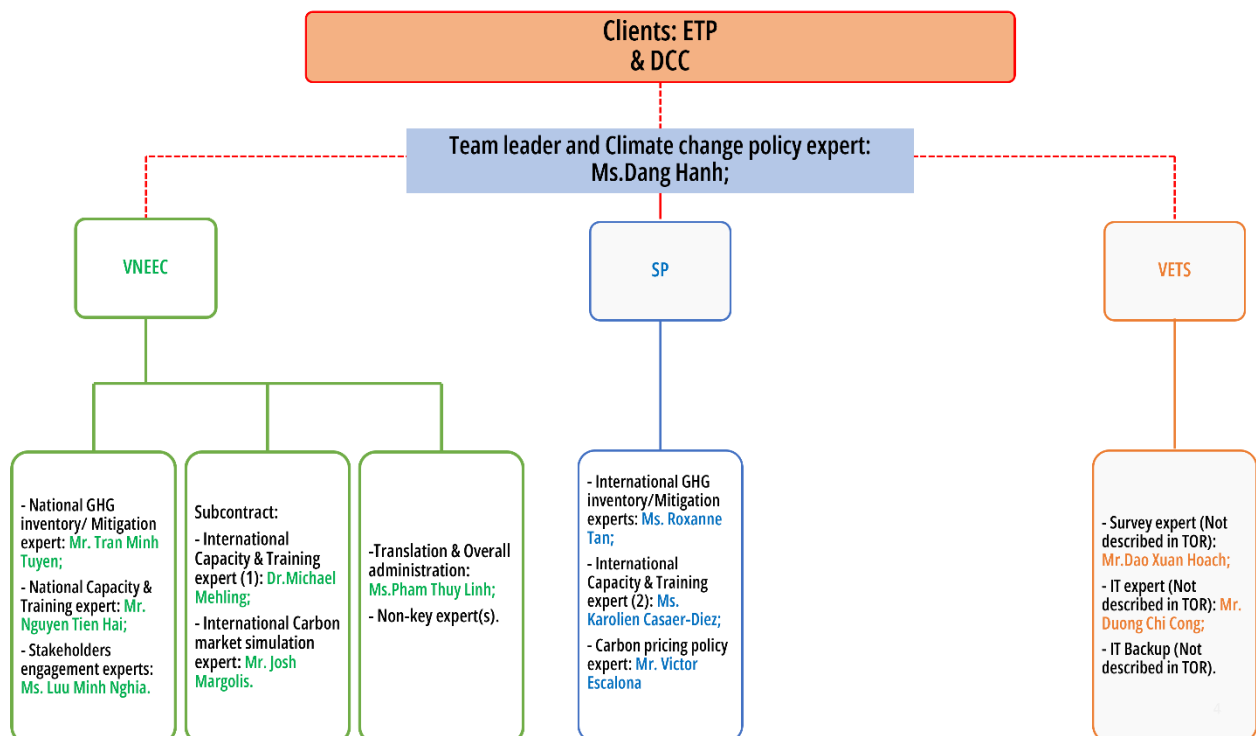


Figure 14: Key project implementation team and interaction with the Client

Source: The Consultant

8 RISKS AND MITIGATION STRATEGIES

During the implementation of the project, it is important to acknowledge that certain issues and challenges may arise, which could potentially impact the outcomes and timeline of the project. Drawing from the Consultant's experience in executing similar projects in Vietnam, a comprehensive risk management strategy has been developed to proactively anticipate, mitigate, and effectively manage these risks. This approach ensures that any potential obstacles are addressed seamlessly, allowing the project to progress smoothly towards its objectives.

Table 14: Identification and management of project risks

Risk	Impact	Mitigation Strategy
Data quality: - Low rate of surveys returned - Poor/incomplete surveys returned	Significant - overall quality	The survey will be carefully designed so as to make it as simple as possible and collect essential data only. Different approaches including support and coordination with government agencies (i.e., Department of Climate Change), online questionnaire forms, follow-up calls and emails, and collaboration with provincial professional associations will be applied to be able to collect data from different sources. The Consultant will facilitate data collection and cooperation with relevant stakeholders for the data collection. Combination with other sources of data available to the Consultant.
Stakeholders' participation expectations	Medium	Close interactions with the stakeholders along the entire assignment to inform them on the goals of the assignment and facilitate their active participation and provision of feedback and inputs.
Insufficient engagement by stakeholders	Significant - overall objectives	Team lead and Stakeholders Engagement Expert will manage the process of engaging the stakeholders and ensuring their advance commitment. The Consultant will develop a strong communications plan, engaging specific stakeholders around their specific interest in the exercise and work with the Client to inform and engage the ministerial responsible persons in facilitating the contacts with key line ministries, industries, and other stakeholders to maintain their engagement and commitment.

Risk	Impact	Mitigation Strategy
		Experts of the team who have a strong network will be mobilized in contacting stakeholders when needed and appropriate.
Unexpected shifts in timeline	Significant - overall the timeline and deliverable	The Team Lead and will closely follow the timeline to be aware of any shifts and deviations from the approved plan. If there is a major change, the Consultant will identify the reason for delays, proposes strategies, corrective activities for unforeseen shifts, including revising the schedule, prioritizing deadlines, and milestones, and communicating with Clients well in advance to seek approval for the potential change in the timeline.

The Consultant will promptly notify the Client of any identified project risks and collaborate to devise a mitigation strategy. This may involve expediting certain activities or securing additional resources to address unforeseen demands. For instance, if there are incomplete survey responses that could impact the overall project objectives, appropriate measures will be taken to ensure their completion. By promptly addressing such risks and implementing suitable mitigation strategies, the project can effectively navigate any challenges and maintain progress towards its goals.

9 THE APPROACH IN GENDER IMPACTS ASSESSMENT

9.1 Gender-sensitive indicators

Within the scale of the project the Consultant will identify the following indicators which will receive financial support promote the participation of female learners to participate in the training.

- **Gender composition of participants:** Track the representation of different genders among participants in the ETS simulation, including stakeholders, decision-makers, and participants in workshops or consultations. This indicator can provide insights into the level of gender inclusivity in the project. The presentation of female participants in workshops and training sessions is targeted to reach at least 40% of the total participants.
- **Participation and engagement:** Assess the level of participation and engagement of different genders in the ETS simulation activities, such as workshops, meetings, and feedback sessions. This indicator can help identify any gender disparities in participation and ensure equal opportunities for involvement. The level of participation and engagement of female participants is targeted to reach 40% participation and engagement of workshops, meetings, and feedback sessions.
- **Capacity building and training effectiveness:** Assess the effectiveness of capacity building and training programs provided as part of the ETS simulation, considering the inclusion of different genders. This indicator can help ensure that gender-specific training needs are met and that all genders have equal opportunities for skill development. The rate of positive evaluation from female participants in training sessions is targeted to reach at least 60% of the female participants.

9.2 Gender-specific needs

In order to foster a safe and inclusive training environment for all participants, irrespective of their gender, the Consultant will take proactive measures to create a welcoming atmosphere. This includes encouraging open dialogue, respecting diverse perspectives, and ensuring that each individual feels comfortable, valued, and respected throughout the entire ETS simulation process. The Consultant is committed to establishing an environment that promotes inclusivity, where participants can freely express themselves and engage in meaningful discussions without any form of discrimination or bias.

In addition, to promote gender inclusivity, the Consultant will tailor support and resources including:

- Gender-sensitive outreach and engagement: Implement outreach strategies that target and engage individuals of all genders, ensuring equal representation and participation in the ETS simulation.
- Recognize and address the unique needs and barriers faced by different genders participating in the ETS simulation. Provide resources, and training opportunities to address gender-specific challenges to actively engage in the simulation.

10 ETP RESULTS BASED MONITORING FRAMEWORK (RBMF)

The results of the Assignment are monitored through the ETP's result-based monitoring framework as described below. All reports will update the achievement of the indicators.

Table 15: ETP's result-based monitoring framework under this Assignment

ETP Pillars	Result(s)	Description	Indicator Number	Program Indicator(s)	Project-level Indicator	Target
Policy alignment with climate commitments	Outcome	1. Strengthened RE and EE policy enabling environment	IN 1-01	No. of new/revised RE and EE policies, laws, regulations, and/or technical standards endorsed and adopted by the national government in coordination with relevant institutions	Not Relevant	
	Outputs	1.1 National RE and EE policies, regulations, standards, and energy plans reflect a clear commitment to Energy Transition agenda and integrated into sectoral plans to contribute to the achievement of Paris Agreement	IN 1.1-01	National energy plans reflect an ambition towards increasing the share of RE/VRE, improving EE, and phasing-out fossil fuels	Not Relevant	
			IN 1.1-02.1	No. of RE and EE policies, laws, regulations, and/or technical standards developed and presented to the government entities	Not Relevant	
			IN 1.1-02.2	No. of RE and EE policies, laws, regulations, and/or technical standards revised and adopted by the government entities		
		1.2 National Fiscal policies, regulations, and Investment policies have undergone reforms to create an Investment Climate that is conducive to investment flow into RE/EE and improves its energy transition readiness for capital and investments	IN 1.2-01	No. of RE and EE related Financing Frameworks and Fiscal reforms developed and presented to the government entities	Policy recommendations for Emission Trading System (ETS) in Vietnam.	1
			IN 1.2-02	No. of Fiscal policy adjustments, Investment framework instruments, established and enacted by the government entities		

		1.3 Energy transition agenda is centrally led and coordinated effectively at a National-level agency/institution that is tasked to champion the cause with right level of authority	IN 1.3-01	Presence of an effective National level agency/institution	Not Relevant	
			IN 1.3-02	Improved dialogue among government ministries and departments for a coordinated response to Energy Transition	Not Relevant	
De-Risking RE and EE Investments	Outcome	2. Increased flow of public and private investments to RE and EE projects in the power and end-user sectors	IN 2-01	Amount of Public, Private, and Blended finance flow into RE and EE	Not Relevant	
	Outputs	2.1 National budgets indicate a resolve to maximize RE/EE capacity by allocating increased amount of public funds and attracting FDI in the RE/EE sector	IN 2.1-01	Amount of Public funding allocated to RE/EE projects	Not Relevant	
			IN 2.1-02	Amount of FDI inflow into RE and EE sector initiatives	Not Relevant	
		2.2 De-risked project finance is accessible via financial institutions generating a pipeline of large-scale RE/EE projects	IN 2.2-01	No. of new and existing, national and international, financing options / instruments de-risked and opened for private and blended financing	Not Relevant	
Extending Smart Grids	Outcome	3. Increasing the amount of RE integrated in smarter grids	IN 3-01	Share of RE/VRE integrated into Smart Grids and improved Smart Infrastructure	Not Relevant	
			IN 3-02	Extent of curtailment reduction	Not Relevant	
	Outputs	3.1 National energy strategy and sectoral plans involve evidence-based planning for an improved national-	IN 3.1-01	No. of technical recommendations and solutions implemented by the grid operators for planning and operation, leading to smart grid	Not Relevant	

		smart-grid system along with related infrastructure and innovative technologies	IN 3.1-02	No. of technical design, demo, modelling projects supported for smart infrastructure	Not Relevant	
Knowledge and Awareness Building	Outcome	4. Increased development of and accessibility to RE/EE knowledge	IN 4-01	Targeted audiences indicate improved knowledge about EE/RE and the ability to sustain Energy Transition efforts	Conducting surveys on readiness and awareness of enterprises and facilities on GHG emission/mitigation and carbon market	80% awareness
	Outputs	4.1. Stakeholders (relevant Government entities, public sector companies, Financial institutions, Private entities, Academia, and Consumers) involved in the RE/EE value chain, are knowledgeable and better informed to advance the energy transition agenda	IN 4.1-01	No. of studies, research, new evidence gathered and published, for raising awareness, improving knowledge base, driving decisions, and dissemination	A report for policy recommendations for Emission Trading System (ETS) in Vietnam.	1
			IN 4.1-02	No. of trainings, knowledge sharing events, and/or awareness workshops organized at national and regional levels building institutional capacity and knowledge networks	1. An inception workshop 2. Organising a series of ETS simulation trainings 3. Planning a carbon market master class for key government officials to learn and exchange governance knowledge and experience of ETS. The participants	

					from Government agencies tentatively include MOF, MONRE, MOIT and State Bank of Vietnam. 4. A final workshop	
			IN 4.1-02 A	Total Number of participants	ETS simulation trainings - at least 200 participants	
			IN 4.1-02 B	Number of Female Participants	At least 30%	
			IN 4.1-02 C	Total no. of Companies / Government bodies supported through Technical Assistance	200 companies	
			IN 4.1-03	No. of articles, press releases on social media, and mass media, for outreach releases on social media, and mass media, for outreach		

The Consultant adheres to rigorous and efficient internal quality check procedures to ensure that all deliverables meet their high standards of quality. Throughout the implementation of the Assignment, the Team Lead will serve as the primary point of contact for the Client, providing continuous support and being readily available for any assistance required. Regular communication via telephone and email will be scheduled between the Team Lead and the Client to provide progress updates and address any issues that may arise.

Besides the result-based monitoring framework, a Quality Assurance Plan encompasses the following elements:

- The services provided by our team will align with the requirements stipulated in the Terms of Reference, ensuring client satisfaction.
- All project deliverables will undergo thorough review by both national and international QA/QC experts before submission.
- Consulting services will incorporate best practices and innovative methods to deliver high-quality outcomes.
- The Consortium's extensive 'in-house' technical and management resources will be leveraged, drawing on their experience and expertise to enhance the tasks and activities.
- A framework for monitoring, evaluation, and response will be established to optimize the services provided throughout the project.

The Consultant will develop a comprehensive project plan that will be mutually agreed upon by all project partners. This plan will serve as a tool to track progress, including activities, milestones, deliverables, key dates (such as Tet holiday and project deadlines), and a matrix of dependencies, inputs, interactions, and critical pathways.

The Consultant has designed the following Quality Assurance Plan to ensure the successful execution of key activities within the Assignment.

Table 16: Quality control plan

No	Task	Assigned expert
1	QC of policies, institutions, financing and stakeholder	Administration & Stakeholder Engagement Experts (QC 1.1); Carbon pricing policy expert (QC 1.2)
2	QC of technical and data aspects	Survey expert (QC 2.1); GHG inventory expert (QC 2.2); Data expert (QC 2.3).
3	QC of training agenda and content	Carbon market simulation expert (QC 3.1); Capacity and training expert (QC 3.2).
4	Overall QA	Team Lead (QA)
5	QA of training agenda and content	International capacity training and carbon market simulation expert(s) (QA training)

Table 17: Specific quality assurance plan for the Assignment

Task/Activity	Quality control measures	Responsibilities
Task 1: Conducting Inception report and Project work plan		QA
a) Activity 1.1: Conducting Inception kick-off meetings	The completeness of the meeting minutes (if any)	QC 1.1
b) Activity 1.2: Composing inception report and updating Work Plan	The completeness and quality of an Inception report and submitted to DCC and ETP	QC 1.1 QC 3.2
Task 2: Organizing an Inception Workshop		QA
a) Activity 2.1: Preparing for the Inception Workshop	Completeness of the draft agenda and list of participants	QC 1.1 QA
b) Activity 2.2: Organizing the Inception Workshop	The completeness of Proceedings of the workshop submitted to DCC and ETP.	QC 1.1
Task 3: Conducting an analysis, surveys and consultation events on readiness and awareness of enterprises and facilities on GHG emission/mitigation and carbon market		QA
a) Activity 3.1: Conducting the desk review and consultations on the ETS (non-emitter entities)	The completeness of the information that was gathered through the desk review	QC 1.1 QC 1.2
b) Activity 3.2: Design the survey questionnaire	The fulfilment of the survey questions in order to assess the readiness and awareness of the selected groups	QC 2.1 QC 3.3
c) Activity 3.3: Determine the scale and coverage of the survey	The number of potential players that will be identified	QC 1.1 QC 2.1
d) Activity 3.4: Data collection, compilation and analysis	The rate and completeness of the responses	QC 3.1 QC 3.2
e) Activity 3.5: Preparing a readiness and awareness assessment report and proposing a list of potential players to participate in the training	The assessment report on awareness, readiness, and capacity training need	QC 1.1 QC 1.2 QC 3.1 QC 3.2
Task 4: Developing a detailed stakeholder mapping and engagement plan to approach and engage the targeted groups		A
a) Activity 4.1: Developing a detailed stakeholder map	The coverage of the stakeholder mapping	C 1.1

Task/Activity	Quality control measures	Responsibilities
b) Activity 4.2: Composing a stakeholder engagement plan	Quality and Completeness of the engagement plan	C 1.1
Task 5: Develop a carbon simulation tool tailor-made for Viet Nam		QA
a) Activity 5.1: Providing and preparing access to and identifying options for adjustment of the carbon simulation tool suitable for use for this Assignment	Ensure the stability and function of the tool will be aligned with the design of the training	QC 2.4 QC 3.1
b) Activity 5.2: Adjusting the CarbonSim tool	Ensure the stability of the tool after adjusting	QC 2.4 QC 3.1
c) Activity 5.3: Testing the tool		
Task 6: Providing a web-based carbon simulation platform		QA
a) Activity 6.1: Developing the web-based platform	The completeness of the platform	QC 2.4
b) Activity 6.2: Developing the content for the web-based platform	Accuracy, consistency and recentness of the content	QC 2.2 QC 3.2 QC 3.3
c) Activity 6.3: Updating the web-based platform	The updating interval of content and project’s activity	QA
Task 7: Organizing a series of ETS simulation training/workshops/forums		QA
a) Activity 7.1: Pre-training activities	The number of potential confirm to participate in the training	QC 1.1
b) Activity 7.2: Organizing the trainings	The completeness of Proceedings of the training submitted to DCC and ETP as well as effectiveness of the training through results assessment.	QA training
Task 8: Planning a master class/study tour for key government officials		QA
a) Activity 8.1: Identify capacity training needs/gaps of government officials	The completeness of results assessments of previous activity which will be used to identify the objectives of the master class	QC 3.2 QC 3.2
b) Activity 8.2: Design the study tour/ master class in close consultation with ETP and DCC	The completeness and comprehensiveness of the proposing list of potential host countries and extended list of countries for the master class	QC 3.2 QC 3.3
Task 9: Conducting a technical report summarizing results of the simulation and impact assessments		QA

Task/Activity	Quality control measures	Responsibilities
a) Activity 9.1: Composing the technical report	Consistency, quality and quantity of insights gathered through the training	QA training
b) Activity 9.2: Organizing the High-level consultations	The completeness of Proceedings of the high-level consultation submitted to DCC and ETP	QA training
Task 10: Conducting final report and final workshops		QA
a) Activity 10.1: Composing the final closing report	Consistency, quality and quantity of insights gathered through the project life cycle	QA training
b) Activity 10.2: Organizing the final workshop	The completeness of Proceedings of the workshop submitted to DCC and ETP.	QC 1.1 QA training

11 ANNEXES

11.1 Annex I: Initial stakeholder list of the Assignment

The lists of stakeholders who are relevant to implementation of this Assignment are provided in the following table.

Table 18: Initial list of stakeholders

No.	Name	Contact Details	Relevance to the Assignment
I	Internal Stakeholders		
1	VNEEC	Ms. Dang Hong Hanh Co-founder & Managing director Email: dhanh@eec.vn Phone: +84 917291417	VNEEC is the lead partner of the Consultant team for the implementation of the project that provides overall coordination and quality assurance of the tasks and deliverables. VNEEC is in charge of the tasks regarding report compilation, stakeholder consultation, data survey analysis, workshop and training organization. VNEEC provides international experts with exceptional expertise and experience in developing/administering the ETS simulation and lecturing on the ETS and carbon markets.
2	VETS	Mr. Dao Xuan Hoach Project manager Email: hoachdx@vets.com.vn Phone: (+84) 969 462 363	VETS is an implementing partner of the Consultant team that provides technical and IT support for the assignment. Specifically, VETS provides technical support and cooperates with VNEEC in conducting GHG accounting and the survey on large emitters, adjusting the carbon simulation tool, and developing a web-based platform for ETS knowledge and training in Vietnam.
3	South Pole	Roxanne Tan Managing Consultant on Climate Policy, Finance and Carbon Markets Email: r.tan@southpole.com Phone: +6583121918	South Pole is an international partner for the implementation of the project and provides quality control for the tasks and deliverables of the assignment. South Pole is in charge of developing a stakeholder mapping and engagement plan, planning a study tour abroad for key government officials. Additionally, South Pole collaborates with VNEEC to review the national data survey analysis and organize a series of ETS simulation training.
4	ETP	Mr. Do Manh Toan National Coordinator Email: toand@unops.org Phone: + 84 96 381 9416	ETP is the Client of the Consultant for this Assignment. ETP supports the transition of Viet Nam toward a greener and low-carbon economy which contributes to the achievement of the UN's SDG TDAs and the Paris Climate Agreement objectives.
II	External Stakeholders		
1	Key line ministries and government authorities		

1.1	Ministry of Natural Resources and Environment (MONRE)	Mr. Le Cong Thanh Deputy Minister Email: lecongthanh@monre.gov.vn	MONRE is the focal line ministry responsible for managing natural resources and the environment, including climate change mitigation action. MONRE also serves as the National Focal Point for the UNFCCC, the Kyoto Protocol, and the Paris Agreement. In addition, MONRE is the coordinating agency for implementing the NDCs. Regarding the development of the domestic carbon market in Viet Nam, MONRE will be in charge of the following specific activities: - In the period by the end of 2027: • Develop regulations on the management of carbon credits, exchange of greenhouse gas emission allowances and carbon credits; develop a regulation on the operation of a carbon credit exchange; • Implement a pilot mechanism for exchanging and offsetting carbon credits in potential sectors; guide the implementation of the domestic and international carbon credit exchange and offset mechanism following the law, regulations and international treaties to which the Socialist Republic of Viet Nam is a signatory; • Establish and organize the pilot carbon credit exchange from 2025; • Implement activities to strengthen capacity and raise awareness about carbon market development. - In the period from 2028: • Operate the official carbon credit exchange in 2028; • Regulate activities to connect and link the domestic carbon market with regional and global carbon markets;
1.1.1	Department of Climate Change (DCC)	Mr. Tang The Cuong Director General Email: ttcuong@monre.gov.vn	According to the assignment from MONRE, DCC is responsible for implement greenhouse gas (GHG) inventory; develop and implement the Measurement, Reporting, and Verification (MRV) system; develop and update a list of sectors and facilities required to conduct GHG inventory; manage and allocate GHG emission allowances; establish mechanisms and policies for carbon pricing; operate the domestic carbon market and connect with the global carbon market; develop

			and implement CTX and offset mechanisms; organize the verification of GHG emission allowances and carbon credits are allowed to trade on the carbon market; monitor the exchange of GHG emission allowances and carbon credits according to the laws and international treaties of the Socialist Republic of Viet Nam as a member.
1.1.1.1	Division of GHG Emission Reduction and Ozone Layer Protection	Mr. Luong Quang Huy Head of Division Mobile: +84 (0)902 870505 Email: huylq98@gmail.com	Division of GHG Emission Reduction and Ozone Layer Protection is a division under MONRE, assigned to: - Assisting DCC Director General with emissions reductions and ozone layer protection issues; - Identifying the emissions benchmark per unit of product that will be applied under the domestic carbon market. Hence, advise and suggest the emissions allowances allocated to regulated entities; - Is the focal point for guiding and implementing the National GHG inventory; implementing the sectoral GHG inventory in the waste sector; - Updating the list of the sector, list of sectors, GHG-emitters subject to greenhouse gas inventory or emitters that must be participated in the carbon market in Viet Nam.
1.1.1.2	Division of Economic and Climate Change Information	Mr. Nguyen Van Minh Head of Division Email: nvminh.dmhcc@gmail.com	Division of Economic and Climate Change Information is a division under MONRE, assigned to: - Assisting DCC Director General in developing mechanisms and policies on carbon pricing; organizing the operation of the domestic carbon market and connecting with the international carbon market; - Developing and managing carbon credits, and carbon offset mechanisms; assisting the Director General in the issuance of certificates to certify projects under the mechanisms under the Paris Agreement on climate change; - Monitoring the implementation of the carbon credit and offset mechanism, the exchange of greenhouse gas emission allowances, and carbon credits in accordance with relevant laws and international treaties. The standing committee of the Joint Committee on the Vietnamese side to implement the joint credit mechanism.
1.1.1.3	Centre of Low-carbon	Mr. Ha Quang Anh Director of Centre	The Centre of Low-carbon Development is under DCC, assigned to conduct GHG inventory, forecast GHG emissions, develop emission

	Development	Email: qanhsilvi@gmail.com	scenarios, evaluate CTX and promote a low-carbon economy.
1.1.2	Viet Nam Environment Protection Fund (VEPF)	Mr. Nguyen Duc Thuan Director Email: ndthuan@monre.gov.vn	VEPF is under MONRE, which implements financial mechanisms and policies for investment projects under the Clean Development Mechanism (CDM), including: - Establish monitoring, management, and collection of fees for the sale of certified emission reduction (CER) certificates - Allocation of support for awareness-raising and promotional activities to enhance understanding of CDM investment projects; review and approval of CDM project documents; management and oversight of CDM projects - Subsidies for products from CDM projects
1.2	The Ministry of Finance (MOF)		MOF is responsible for establishing the carbon trade exchange and promulgating finance administration mechanisms for activities of the carbon market, including setting the price of carbon emission allowances and managing the revenue generated from the carbon market.
1.2.1	Department of Legal Affairs	Ms. To Nguyen Cam Anh Phone: + 84 987568792 Email: tonguyencamanh@mof.gov.vn	The Legal Department is a unit under MOF, responsible for coordinating or participating in developing financial mechanisms, policies, projects, and legal normative documents assigned by MOF.
1.2.2	State Securities Commission of Viet Nam (SSC)	Ms. Vu Thi Chan Phuong Chairman Phone: (+84) 2439340750	State Securities Commission is under MOF, which can play a significant role in promoting sustainable financial mechanisms such as green bonds and encouraging businesses to participate in initiatives aimed at mitigating GHG. Specifically, SSC can participate in the development and proposal of policies and regulations related to the carbon trade exchange of Vietnam.
1.3	The Ministry of Industry and Trade (MOIT)	Mr. Hoang Van Tam Deputy Head of Climate Change and Green Growth Office, Department of Energy Efficiency and Sustainable Development Email: tamhv@moit.gov.vn Phone: +84 913 599 409	MOIT is responsible for promoting the development of low-carbon and climate-resilient industries. The MOIT's expertise in the industrial sector, as well as its close relationships with businesses, will be critical to the smooth operation of the ETS in this sector. The ETS is a market-based mechanism that can help to incentivize businesses to reduce their emissions in this sector. MOIT is also responsible for ensuring that the carbon market does not have a negative impact on the competitiveness of Vietnamese businesses.
1.4	Ministry of Planning	Mr. Nguyen Chi Dung	MPI is responsible for developing the national climate change strategy and for coordinating with

	and Investment (MPI)	Minister Email: nguyenchidzung@mpi.gov.vn	other ministries on climate change issues. The MPI's expertise in planning and investment, as well as its close relationships with other government agencies, will be critical to the smooth operation of the ETS.
1.5	Ministry of Agriculture and Rural Development (MARD)	Mr. Le Minh Hoan Minister Email: leminhhoan@mard.gov.vn	MPI is responsible for developing the national climate change strategy and for coordinating with other ministries on climate change issues. The MPI's expertise in planning and investment, as well as its close relationships with other government agencies, will be critical to the smooth operation of the ETS.
1.6	Ministry of Transport (MOT)	Mr. Tran Anh Duong Director, Department of Environment Phone: +84 903 269 542 Email: trananhduong@mt.gov.vn	MOT is responsible for promoting the development of low-carbon transport.
1.7	Viet Nam Chamber of Commerce and Industry (VCCI)	Mr. Pham Tan Cong Chairman and President Phone: (+84) 24 35742022	VCCI is representing the interests of businesses in the carbon market.
2	Large GHG emitters		
2.1	Emitters under the management of the industry and trade sector	Quang Ninh Thermal Power Joint Stock Company, Samsung Display Viet Nam CO., LTD., Hai Phong Thermal Power SJC, etc.	Out of the total 1912 entities required to conduct GHG inventory, this sector accounts for 1662 enterprises, accounting for more than 80% of enterprises. The biggest emitters in the Energy Production Industry, Coal Mining, and Chemical Production sectors are under this sector. Therefore, they are the key stakeholders and potential participants of the training sessions on ETS.
2.2	Emitters under the management of the transportation sector	Viet Nam Petroleum Transport JSC, Trung Nghia Transport Services And Investment Joint Stock Company, etc.	The transportation sector has 70 enterprises required to conduct greenhouse gas inventory. The businesses with the highest levels of TOE are primarily located in the northern region.
2.3	Emitters under the management of the construction sector	Song Thao Cement Factory of Vicem Song Thao Cement Joint Stock Company; Cam Pha Cement Factory of Cam Pha Cement	There are 104 businesses in the construction sector that are mandated to conduct GHG inventory. Among them, the enterprises in the cement production industry have the highest emissions.

		Joint Stock Company, etc.	
2.4	Emitters under the management of the natural resources sector	Nam Son Waste Treatment Complex of Hanoi Urban Environment Company, Phuoc Hiep Solid Waste Treatment Complex, etc.	While there are 76 businesses mandated to conduct greenhouse gas inventory, these enterprises have high emissions. Enterprises in this sector mostly are state-owned enterprises.
3	Financial institutions, and banks	Viet Nam Bank for Social Policies, Viet Nam Bank for Agriculture and Rural Development, etc.	Financial institutions and banks act as intermediaries, connecting buyers and sellers in the carbon market to ensure smooth and efficient transactions; and offer loans and financial products to businesses and industries looking to implement GHG mitigation measures.
4	Media	VTV, VnExpress, Viet Nam News Agency (VNA), etc.	The media plays an important role in the development of the ETS in Vietnam. The media can help in: Raising public awareness of the ETS and its benefits. Explaining the ETS to businesses and the public. Holding the government accountable for the implementation of the ETS. Monitoring the performance of the ETS and identify any problems. The media can also help in promoting the ETS as a market-based mechanism that can help to incentivize businesses to reduce their emissions. By raising public awareness of the ETS and its benefits, the media can help to create a more favorable environment for the implementation of the scheme.
5	Multilateral and bilateral donors		Multilateral and bilateral donors provide support to aid Viet Nam in creating and managing carbon markets and ETS. They assist Viet Nam in assessing Vietnam's readiness and identifying the gaps, thus, providing support to help Viet Nam realize the carbon market, but there is a specific difference between the two types of donors.
5.1	Multilateral donors	World Bank, ETP, UNDP, etc.	Multilateral donors are typically international organizations involving multiple countries working together, therefore, they provide support on regional and global scales. The roles and contact details of multilateral donors will be specified in the donor map and donor coordination strategy.
5.2	Bilateral donors	USAID, JICA, GIZ, etc.	Bilateral donors are individual countries. They provide assistance to Viet Nam based on their bilateral agreements or partnerships. The roles and contact details of multilateral donors will be

			specified in the donor map and donor coordination strategy.
6	ETS service providers and offset developers	To be specified in the detailed stakeholder engagement plan	They are the third parties to provide services related to developing carbon credit projects, GHG inventory, abatement, and supporting the effective operation of ETS.

11.2 Annex II: Initial stakeholder engagement plan outline

11.2.1 Introduction

[The Stakeholder Engagement Plan (SEP) is prepared by the Consultant as an integral component of the Assignment. The purpose of the SEP is to provide a comprehensive framework for stakeholder engagement during the preparation stage. It is important to note that the SEP is a dynamic document that will undergo continuous updates and enhancements by the Consultant, in collaboration with the ETP and DCC, throughout the project's life cycle.]

11.2.2 Stakeholder identification and approach

[The general approaches to engage the stakeholders may include:

- *For Government Agencies and Regulatory Bodies:*
 - ❖ *Conduct regular meetings with representatives from relevant agencies to discuss project updates, progress, and regulatory compliance.*
 - ❖ *Share project documentation, including reports and analyses, to ensure transparency and gather feedback on the content of the training.*
 - ❖ *Seek their input in shaping the simulation framework and aligning it with existing emissions regulations.*
- *For industry representatives:*
 - ❖ *Organize industry-specific roundtable discussions to understand their perspectives on emissions trading and its potential impacts.*
 - ❖ *Collaborate with industry associations to disseminate information about the simulation and encourage participation.*
 - ❖ *Establish working groups with industry representatives to address sector-specific challenges and develop tailored solutions.*
- *Financial Institutions:*
 - ❖ *Arrange meetings with representatives from financial institutions, including banks, investment firms, and insurance companies, to discuss the role of finance in supporting emissions trading initiatives.*
 - ❖ *Provide educational sessions on the financial aspects of emissions trading, including carbon pricing mechanisms, investment opportunities, and risk management strategies.*
 - ❖ *Collaborate with financial institutions to develop specialized financial products and services that support emissions trading, such as carbon offset credits or green investment funds.*
- *For service provider:*

- ❖ *Identify potential offset providers and establish communication channels to introduce the emissions trading simulation project and its objectives.*
- ❖ *Provide information sessions for offset providers to provide an overview of the project, its requirements, and the opportunities for their participation.*
- ❖ *Share comprehensive documentation, including guidelines and technical specifications, to ensure offset providers have a clear understanding of the simulation process and expectations.*
- *NGOs/NPOs and other stakeholders:*
 - ❖ *Provide educational materials and resources to NGOs and NPOs to enhance their understanding of emissions trading concepts, mechanisms, and potential benefits.*
 - ❖ *Organize a roundtable with NGOs and NPOs, focusing on the role they can play in advocating for environmental integrity and social equity within emissions trading.*
 - ❖ *Collaborate with NGOs and NPOs to develop and disseminate information materials targeted at the broader public, raising awareness about the simulation and its objectives.]*

11.2.3 Propose strategies for information disclosure

[The strategies for information disclosure may include:

- *Information category:*
 - ❖ *Project Overview: Provide a clear and concise description of the emissions trading simulation project, including its objectives, scope, and expected outcomes.*
 - ❖ *Program Updates: Regularly share updates on project progress, milestones achieved, and upcoming activities to keep stakeholders informed and engaged.*
- *Frequency of Disclosure:*
 - ❖ *Regular Updates: Provide consistent and timely updates on project progress, market trends, and policy developments to maintain stakeholders' interest and awareness.*
 - ❖ *Milestone Announcements: Communicate significant milestones, such as the launch of the simulation, forthcoming course, or successful completion of key activities, to celebrate achievements and maintain stakeholder engagement.*
 - ❖ *Ad hoc Communications: Respond promptly to emerging issues, stakeholder queries, or concerns, demonstrating a commitment to open dialogue.]*

11.2.4 Communication plan/strategy

Will be further described in the communication plan in Annex III below.

11.2.5 Review comments and feedbacks

[The review of comments and feedback is a critical part of stakeholder engagement in the Assignment. It involves gathering feedback through various channels, such as online platforms, email communication, workshops, and surveys. Once collected, the feedback is subjected to a thorough review process, which includes categorization and analysis by the Consultant.

The input of relevant project team members, technical experts, and stakeholders is sought to ensure diverse perspectives are considered. The feedback received informs decision-making, with

a focus on aligning decisions with project goals and stakeholder interests. The final decisions and a summary of how comments were taken into account are reported back to stakeholders through concise documents, project updates, newsletters, and the project website.

The engagement of stakeholders through feedback contributes to the continuous improvement of the simulation by integrating relevant suggestions and fostering a culture of learning. The establishment of a feedback loop ensures stakeholders are informed of the impact their input has on project decisions and outcomes, promoting a sense of inclusivity in the simulation's development.]

11.2.6 Management functions and responsibilities

[The integration of stakeholder engagement activities within the Assignment's management system will span across all phases, ranging from the preparatory stage to the finalization phase. Each team member will actively incorporate the SEP into their respective tasks, working closely under the guidance and supervision of the Stakeholder engagement expert. It is imperative that every activity undertaken within this project adheres closely to the predetermined stakeholder criteria outlined earlier.]

11.3 Annex III: Initial communication plan

11.3.1 About the Communication Plan

The Communication Plan of this Assignment will establish a comprehensive framework for effective communication and support the successful implementation of the project. This document serves as a centralized mechanism for facilitating communication among all identified key stakeholders involved in the Assignment.

The primary objective of the Communication Plan is to harmonize stakeholder needs and establish standardized communication channels that facilitate the exchange of project-related information. By implementing these channels, the plan enables efficient processes for disseminating project updates, conveying the current status, addressing issues, and providing a platform for stakeholders to provide feedback.

Throughout the lifecycle of the Assignment, various types of information will be communicated to the stakeholders. These include details regarding the execution of the project, the status of implementation, general awareness information about the Assignment, and other pertinent information relevant to the stakeholders' interests.

Stakeholders have been categorized into two broad groups: internal stakeholders and external stakeholders. The internal stakeholders encompass the implementing partners and the donor, who are directly involved in managing and executing the Assignment. On the other hand, external stakeholders consist of government authorities, targeted survey participants, and other relevant entities who possess a vested interest in the Assignment and may have a direct or indirect influence over its outcomes. It is crucial to engage with these stakeholders through consultation and effective communication.

To ensure streamlined and targeted communication, detailed communication matrices have been developed. These matrices match specific stakeholders with the appropriate types of information and communication methods. Additionally, the Communication Plan outlines the frequency and media channels to be utilized for each type of communication. These matrices serve as the

foundation for the Consultant team's communication strategy, answering the key questions of who, what, where, when, why, and how communication will take place with project stakeholders.

By implementing this robust Communication Plan, the Assignment aims to foster transparency, enhance stakeholder engagement, and facilitate effective two-way communication throughout the project's duration.

11.3.2 Communications goal and objectives

The Communication Plan aims to facilitate effective communication among implementing partners, the Consultant team, the donor, and other relevant stakeholders. The plan encompasses the following goals:

- Create awareness and understanding: Increase organizational awareness of the Assignment's potential impact.
- Convey implementation status: Clearly communicate the progress of the project and coordinate efforts.
- Address stakeholder issues: Promptly respond to concerns and address the needs of key stakeholders.
- Foster continuous improvement: Incorporate suggestions and recommendations to enhance the project's processes.
- Report implementation problems: Communicate any issues that may affect the Assignment's implementation and their potential impact.
- Ensure stakeholder awareness: Make sure that employees of implementing partners, external stakeholders, and other interested parties are aware of the Assignment's existence and its rationale.
- Maintain enthusiasm: Reinforce the value of the Assignment and sustain enthusiasm among stakeholders.

The objective of this Communication Plan is to support the Consultant team by:

- Providing guidelines: Offer implementing partners and team members a framework to communicate about timelines, resource requirements, and significant plan changes.
- Establishing a protocol: Define a protocol for communicating with various stakeholders, such as employees, government agencies, and survey participants, to highlight the benefits and necessity of their cooperation in the Assignment's initiatives.
- Sharing communication channels: Provide information on appropriate communication channels and contact points for effective engagement.

11.3.3 Execution of the Assignment

The information pertaining to the day-to-day project procedures, including:

- The survey and consultations with key stakeholders (see Tasks 2, 3, 7, 8 and 9)
- Organizational structure, Management control and project staff (see Section VI)
- Risks and mitigation strategies (see Section VII)
- Work plan (see Section 11)

Generic information

Background information and other related information about the Assignment are provided in Section I.

Level of information/communication

To ensure effective communication and engagement, it is crucial to provide implementing partners' employees and other relevant stakeholders with project details that align with their level of involvement and the impact on their work. Two levels of information have been defined: **"For Your Information" (FYI)** and **"For Your Action" (FYA)**. This ensures that stakeholders receive the appropriate level of detail based on their roles and responsibilities within the project. From top management to front-line associates, each stakeholder will receive the information necessary to either stay informed about the project's progress (FYI) or take specific actions based on the provided information (FYA). This approach aims to facilitate clear and targeted communication that supports effective collaboration and decision-making throughout the project lifecycle.

Frequency and media options for communications

The following key identifies the various frequency and media options for communications.

Table 19: Frequency and media options for communications

Communications Frequency	Communications Media
Daily	Consultation workshops
BI-Weekly	Consultation meetings
Weekly	Meetings
Monthly	Training
Quarterly	Virtual meeting
Upon Assignment milestones	Telephone calls
As needed	Official letter/documents sending by post
As requested	Documents / Presentations
Ad hoc	E-mail

Communication matrices

Three distinct communication matrices have been developed to facilitate effective communication within the implementing partners of the Consultant team and between the Consultant team and the two main groups of stakeholders. These matrices serve the purpose of aligning specific types of information with the appropriate stakeholders involved. Additionally, the matrices outline the recommended frequency and medium of communication for each stakeholder group. The dynamic nature of the matrix format allows for flexibility in accommodating changes in the information shared, its frequency, and the preferred format.

A significant portion of the communicated information is focused on internal communication within the Consultant team. This emphasis stems from the recognition that a well-informed team is better equipped to effectively convey the strategy, goals, objectives, and status of the Assignment's efforts. Furthermore, as the majority of the communication originates from within the team, establishing effective communication channels and practices internally contributes significantly to the success of the overall communication efforts with external stakeholders.

Stakeholders: The Consultant

Table 20: Communication within the Consultant

Topic	Information	Venue / Media	Frequency	Purpose
Project Execution / Status	The Consultant team accomplishments, project progress, issues and challenges	Implementing partner Lead Meeting / Individual Team Meeting	BI-Weekly	FYA
Status	High-level project status / progress (Weekly Highlights)	E-mail	Weekly	FYI / FYA
Project Execution	Issues and challenges	Team meeting	Weekly	FYA
Status	Detailed project progress	Status Report (written document)	BI-Weekly	FYI / FYA
Status	Overview of project progress	Monthly Extended Team Status Meeting / e-mail	Monthly	FYA / FYI
Status	High-level project progress	All Hands Meeting	Quarterly	FYI
Generic Information / Status	Popular Press Articles on the industry progress related to the Assignment	E-mail, implementing partners' websites	Ongoing	FYI

Stakeholders: The Consultant team and ETP, DCC

Table 21: Communication between the Consultant and ETP, DCC

Topic	Information	Venue / Media	Frequency	Purpose
Status	High-level project status / progress	E-mail	Monthly/ As requested	FYI / FYA
Project Execution	Support in coordination and facilitation of the Assignment	Official letter/ Meeting/ E-mails	As needed (for data collection/ working with government authorities)	FYI/FYA
Project Execution	Review project focus and gain guidance from team	Meeting/ Virtual meeting	Monthly/ As requested – Specify day/time	FYI / FYA
Status	Confirm direction; cultivate sponsorship and report status, findings and issues.	Meeting/ Virtual meeting	As requested, – Specify day/time	FYI / FYA
Generic Information / Status	Popular press articles/ important information related to the Assignment	E-mail	On going	FYI

Stakeholders: The Consultant team, companies, and enterprises listed in Decision 01/2022/QD-TTg, industrial associations, banking and financial institutions, related donors working in the fields, and others.

Table 22: Communication between the Consultant and private sector

Topic	Information	Venue / Media	Frequency	Purpose
Data collection	Needed data to assess the awareness and readiness regarding the carbon market	Official letter/documents sending by post, E-mail	Ongoing (during data collection period)	FYI/FYA
Data collection	Status on filling questionnaires, sending back required information	Telephone calls/ E-mails / Texting apps	Ongoing (during data collection period)	FYI/FYA
Awareness	Status and upcoming events for consultation	E-mails	As needed (consultation meetings/ workshop)	FYI
Consultation	Expert opinions/ perspectives and technical information	Consultation workshop/ Consultation meeting/ Virtual meeting/ E-mails/ Telephone calls	As needed/ as per consultation workshops required under the Assignment	FYI
Consultation	Interest/affected stakeholder group's opinions/ perspectives/ concerns over topics of the Assignment	Consultation workshop/ Consultation meeting/ Virtual meeting/ E-mails/ Telephone calls	As needed/ as per consultation workshops required under the Assignment	FYI
Awareness	Questions related to the Assignment	Hotline / e-mail	Ongoing	FYI
Generic Information / Status	Popular Press Articles on the industry progress related to the Assignment	Web Page / e-mail	Ongoing	FYI

11.4 Annex IV: Kick-off meeting with ETP

ETS Piloting and Simulation in Vietnam Minutes of project kick-off meeting with ETP

BACKGROUND

Objectives:

- To introduce members of the Consortium from Consultant and ETP team;
- To give an overview of the project
- To discuss and get consensus about issues related to the project

Place: Online via Google meeting

Time: 4 PM – 5 PM (ICT), 31 July 2023

PARTICIPANTS

ETP team: John Robert Cotton, Masako Nemoto, Praewpan Intapanya, Toan Manh Do

Consultant team:

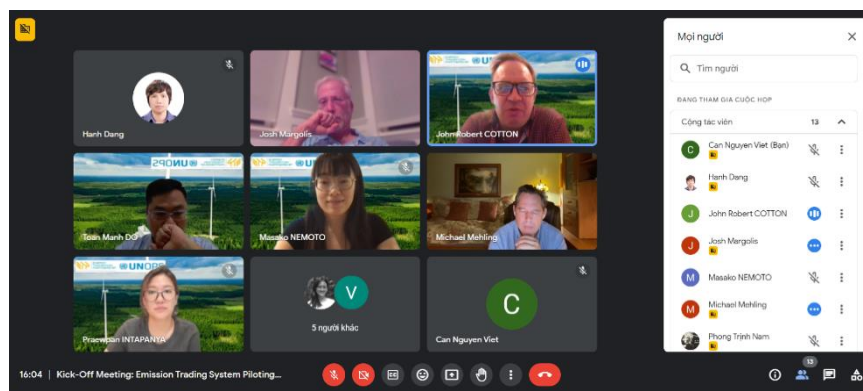
VNEEC: Dang Hong Hanh, Josh Margolis, Michael Mehling, Nguyen Viet Can, Pham Thuy Linh, Dang Tra My, Hoang Thuy An;

South Pole: Roxanne Tan;

VETS: Dao Xuan Hoach, Tram Anh.

AGENDA

Time	Content	Presented by
16:00 – 16:05	Introduction of ETP team	ETP team
16:05 – 16:10	Introduction of the Consultant team	Consultant team
16:10 – 16:25	Project timeline, methodology and approach to the survey	VNEEC team
16:25 – 16:35	Overview of the carbon market simulation tool (CarbonSim)	VNEEC team (Josh)
16:35 – 16:55	Q&A and discussion	All participants
16:55 – 17:00	Conclusion	ETP team



Participants joined the meeting via Google meeting

CONTENTS

1. Members of the ETP team and Consultant team take turns introducing themselves and their roles and responsibilities under the assignment

2. Project timeline, methodology and approach to the survey

Ms. Hanh – VNEEC presented the overview of the project, timeline, team management structure, and general approach for implementation of Task 3.

- A short introduction on the Consortium structure, and responsibilities of VNEEC and VNEEC's sub-contractors (South Pole, Vets) for each task under the project; and the timeline for implementation
- highlighting the importance of not only working closely with relevant stakeholders and government authorities but also cooperating with media agencies to make widespread dissemination of the project's impact
- Clarifying the methodology and approach to the survey in task 3 “Conducting an analysis, surveys, and consultation events on readiness and awareness of enterprises and facilities on GHG emission/mitigation and carbon market”

3. Overview of the carbon market simulation tool (CarbonSim)

Josh – VNEEC presented a short introduction of the carbon market simulation tool (CarbonSim): schedule, objectives, benefits, and calibration of the tool and how the tool can be used for the training sessions under the project.

□ Participants expressed interest in tool and scale up utilization of the tool in similar training projects in the Philippines and Indonesia.

4. Discussion

Comments on the presentations:

- **Toan – ETP:** Project team should reschedule the inception workshop from its initial dates of 15-16 August to early September. This adjustment will help stakeholders thoroughly review the inception report before the inception workshop takes place.

- **Toan – ETP:** ETP is currently supporting the Ministry of Finance (MOF) in assessing Vietnam's readiness and international experience for carbon trade exchange design. The project aims to identify the specific capacity building needs of government authorities from successful ETS implementations in other countries. Thus, he suggested the project team should collaborate closely with MOF to gain a comprehensive understanding of what government officers want to learn about ETS and the necessary actions to be taken soon. Additionally, a UK funder is interested in funding a study tour for government officers to learn about ETS implementation and design in the UK. Therefore, Task 8 should be conducted right after the Inception workshop, preferably getting the proposal in October and planning for the study tour in November 2023, before the holiday season in the UK.

Next steps:

- The first draft of the inception report will be delivered on 18 August 2023
- Proposal for the inception workshop for endorsement of both ETP and DCC.

12 REFERENCES

Bộ Tài nguyên và Môi trường. (2022, October 15). *Quy định kỹ thuật đo đạc, báo cáo, thẩm định giảm nhẹ phát thải khí nhà kính và kiểm kê khí nhà kính lĩnh vực quản lý chất thải*.

<https://monre.gov.vn/Pages/quy-dinh-ky-thuat-do-dac,-bao-cao,-tham-dinh-giam-nhe-phat-thai-khi-nha-kinh-va-kiem-ke-khi-nha-kinh-linh-vuc-quan-ly-chat-thai.aspx>

Environmental Defend Fund. (2021). *What Environmental Market Experts and Educators Say About CarbonSim*.

ICAP. (2021). *ICAP ETS Briefs*.

World Bank. (2023). *State and Trends of Carbon Pricing 2023*.

<https://openknowledge.worldbank.org/entities/publication/58f2a409-9bb7-4ee6-899d-be47835c838f>